

Computer Networks

BCSE308L

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Course Outcomes

1. Interpret the different building blocks of Communication network and its architecture.
 2. Contrast different types of switching networks and analyze the performance of network
 3. Identify and analyze error and flow control mechanisms in data link layer.
 4. Design sub-netting and analyze the performance of network layer with various routing protocols.
 5. Compare various congestion control mechanisms and identify appropriate transport layer protocol for real time applications with appropriate security mechanism.
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Module – 1:

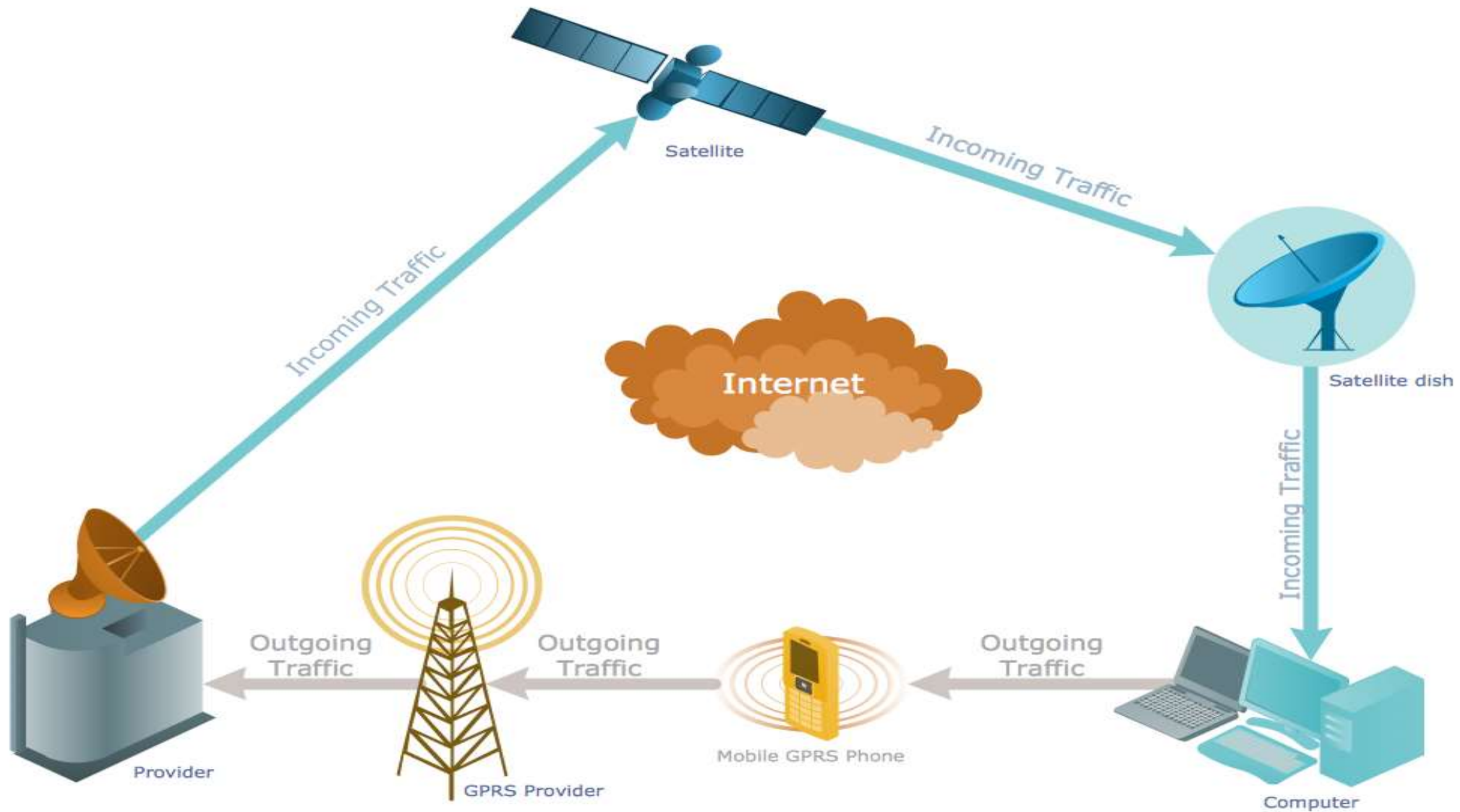
Networking Principles and Layered Architecture

Data Communications and Networking:

A Communications Model – Data Communications - Evolution of network, Requirements , Applications, Network Topology (Line configuration, Data Flow), Protocols and Standards, Network Models (OSI, TCP/IP)

Introduction

- Computer networking is central to today's constant evolution of the information technology (IT) landscape. Network and communication technologies have been influential in this rise.
- Computer networking makes the interconnection of endpoints and devices possible on local area networks (LANs) or wide area networks (WANs). This enables the interaction, communication, and sharing of resources.
- A network additionally enables associated PCs to share documents and information and also equipment assets, i.e., scanners, plotters, projectors, and storage devices, making it simpler to gather and administer data, and enabling clients to work together. The system was simply not advanced in one day; rather took a long time to be an all the more incredible, productive, and dependable system.



History

Year / Period	Milestone	Key People / Organizations	Significance
1950s	Early computer communication research	Military & academic researchers	Laid the groundwork for connecting computers.
1962	Packet switching concept proposed	J. C. R. Licklider; Paul Baran	Introduced resilient digital communication concepts.
1965	First wide-area computer connection	Lawrence G. Roberts; Thomas Merrill	Demonstrated remote computer communication.
1969	ARPANET launched	ARPA	First operational packet-switched network.
1971	Email introduced	Ray Tomlinson	Popularized network communication.
1973–1974	TCP designed	Vint Cerf; Bob Kahn	Foundation of internetworking.
1978	TCP split into TCP and IP	Internet research community	Improved networking architecture.
1983	TCP/IP adopted by ARPANET	ARPANET	Birth of the modern Internet.

1984	Domain Name System (DNS) introduced	Paul Mockapetris	Simplified Internet addressing.
1986	NSFNET established	National Science Foundation	Expanded academic networking.
1989–1991	World Wide Web invented	Tim Berners-Lee	Enabled hyperlinked web pages.
1993	Mosaic web browser released	NCSA	Accelerated Web adoption.
Mid-1990s	Commercial Internet expansion	Internet Service Providers	Brought the Internet to the public.
1997	Wi-Fi (IEEE 802.11)	IEEE	Enabled wireless networking.
1998	Google founded	Google	Improved web information access.
2000s	Broadband expansion	Telecom providers	Delivered high-speed Internet.
2006	Cloud computing growth	Amazon Web Services	Expanded cloud-based networking.
2010s	4G mobile Internet	Telecom industry	Enabled high-speed mobile connectivity.
2019 onward	5G deployment	Global telecom industry	Higher speed, lower latency, IoT support.
2020s	Edge computing, AI networking, Wi-Fi 6/6E/7	Technology industry	Supports AI and real-time applications.

Comparative Summary

Era	Access Speed	Key Tech	Dominant Use Case
1995-2000	Kbps	Dial-up, Ethernet	Basic web, email
2000-2010	Mbps	Wi-Fi, 3G, VPN	Mobile browsing, VoIP
2010-2020	100+ Mbps	4G, SDN, Cloud, IoT	Streaming, cloud, IoT
2020-2025	Gbps	5G, AI, SASE, Wi-Fi 6	Real-time apps, AI, AR/VR

DATA COMMUNICATIONS

When we communicate, we are sharing information. This sharing can be local or remote. Between individuals, local communication usually occurs face to face.

Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable. For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).

1. Delivery. The system must deliver data to the correct destination.

2. Accuracy. The system must deliver the data accurately.

3. Timeliness. The system must deliver data in a timely manner.

4. Jitter. Jitter refers to the variation in the packet arrival time.

Five components of data communication

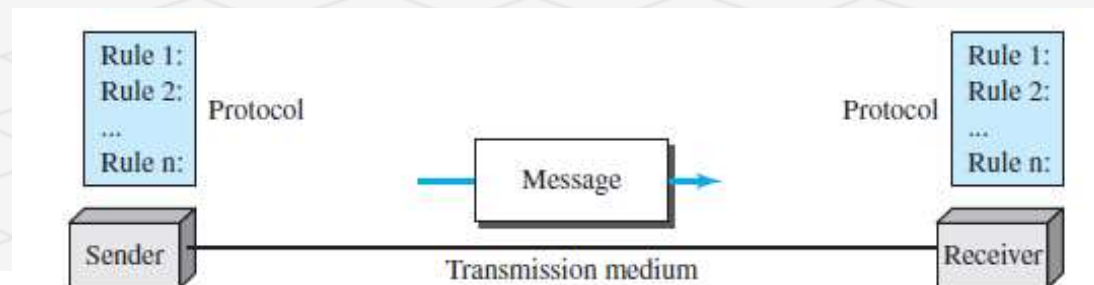
1. Message

2. Sender

3. Receiver

4. Transmission medium

5. Protocol



Data Representation

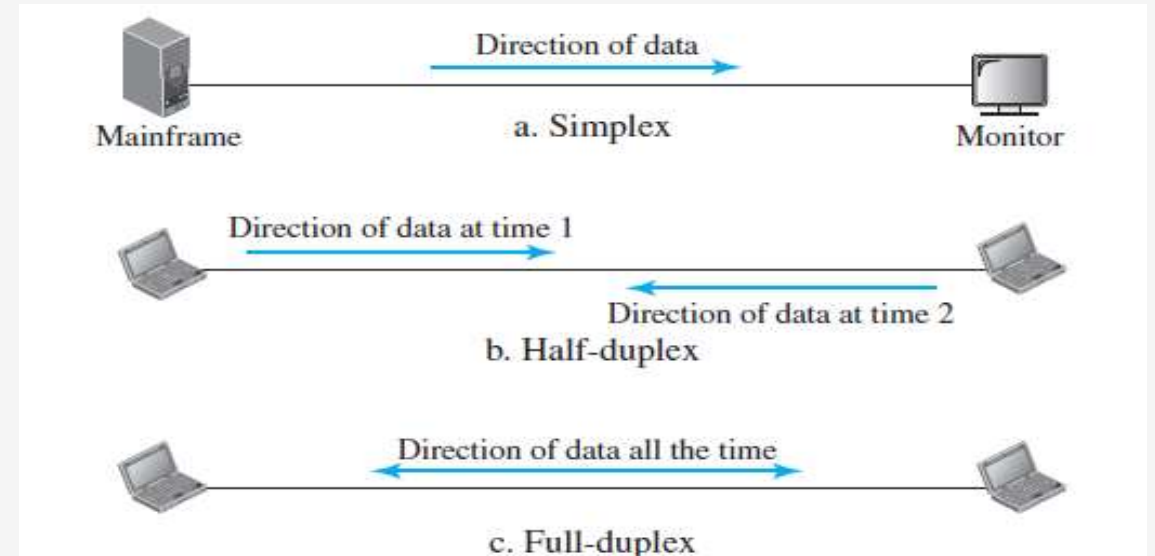
Data Type	Representation	Description	Example
Text	ASCII, Unicode (UTF-8, UTF-16)	Characters are represented using binary codes.	Letter A = 65 (ASCII)
Numbers	Binary	Decimal numbers are converted into binary format for processing.	25 = 11001 ₂
Images	Pixels and binary values	Images are stored as a collection of pixels, with each pixel represented in binary.	JPEG, PNG
Audio	Digital samples	Sound is converted into binary using sampling and quantization.	MP3, WAV
Video	Sequence of image frames with audio	Videos consist of multiple image frames and audio encoded in binary.	MP4, AVI

Data Flow

Data flow in computer networks refers to the **direction in which data is transmitted between two or more devices**.

It determines how communication occurs between the sender and the receiver. Depending on the communication requirements, data flow can be classified into three types:

- a) In **simplex communication**, data flows in **only one direction**. One device always acts as the sender, and the other always acts as the receiver. The receiving device cannot send data back.
- b) In **half-duplex communication**, data can flow in **both directions**, but **not at the same time**. Each device can send and receive data, but they must take turns.
- c) In **full-duplex communication**, data flows in **both directions simultaneously**. Both devices can send and receive data at the same time.



Networks

A **network** is the interconnection of a set of devices capable of communication.

Devices in a network can be:

Host (End System): Large computer, desktop, laptop, workstation, mobile phone, security system.

Connecting Devices: Router, switch, modem, etc.

Devices are connected using:

Wired transmission media: Twisted-pair cable, coaxial cable, fiber-optic cable.

Wireless transmission media: Air (radio waves, infrared, satellite).

Example: Connecting two computers using a home router creates a small network.

Network Criteria

A good network should satisfy three important criteria:

1. Performance

Performance measures how efficiently a network transfers data.

Performance Metrics

Transit Time: Time taken for a message to travel from the sender to the receiver.

Response Time: Time between sending a request and receiving a response.

Networking Metrics

Throughput: Amount of data successfully transferred in a given time.

Delay (Latency): Time taken for data to travel across the network.

Network Criteria

2. Reliability

Reliability refers to the ability of a network to perform consistently without failure.

Reliability is measured by:

Frequency of failures

Recovery time after failure

Ability to withstand disasters (Robustness)

3. Security

Security protects the network and its data.

Security includes:

Protecting data from unauthorized access

Protecting data from damage or modification

Implementing recovery policies after attacks or data loss

Physical Structures

Type of Connection

A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another. For visualization purposes, it is simplest to imagine any link as a line drawn between two points.

1. Point-to-Point Connection

A dedicated communication link between **two devices**. The entire bandwidth is reserved for these two devices only.

Characteristics

Connects only two devices.

High speed and secure.

Can use wired or wireless media.

Physical Structures

2. Multipoint Connection (Multidrop)

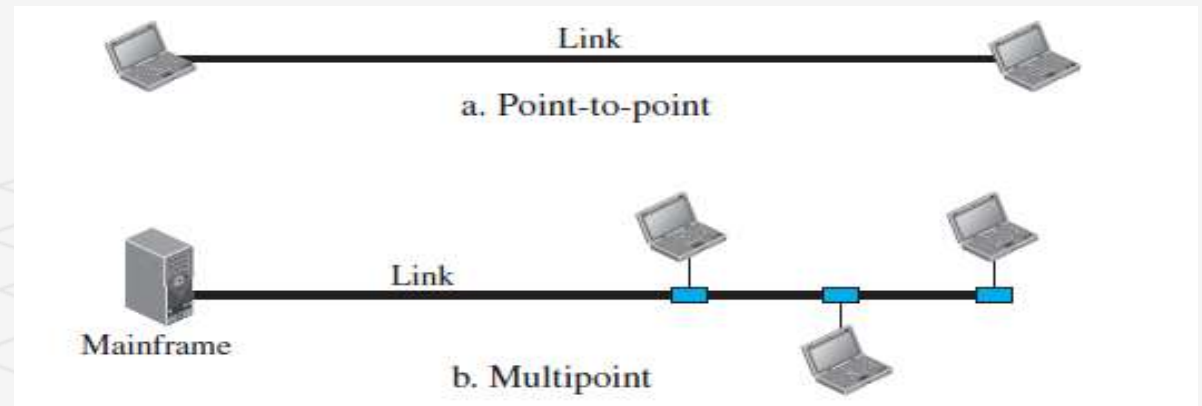
A single communication link is shared by **more than two devices**.

Characteristics

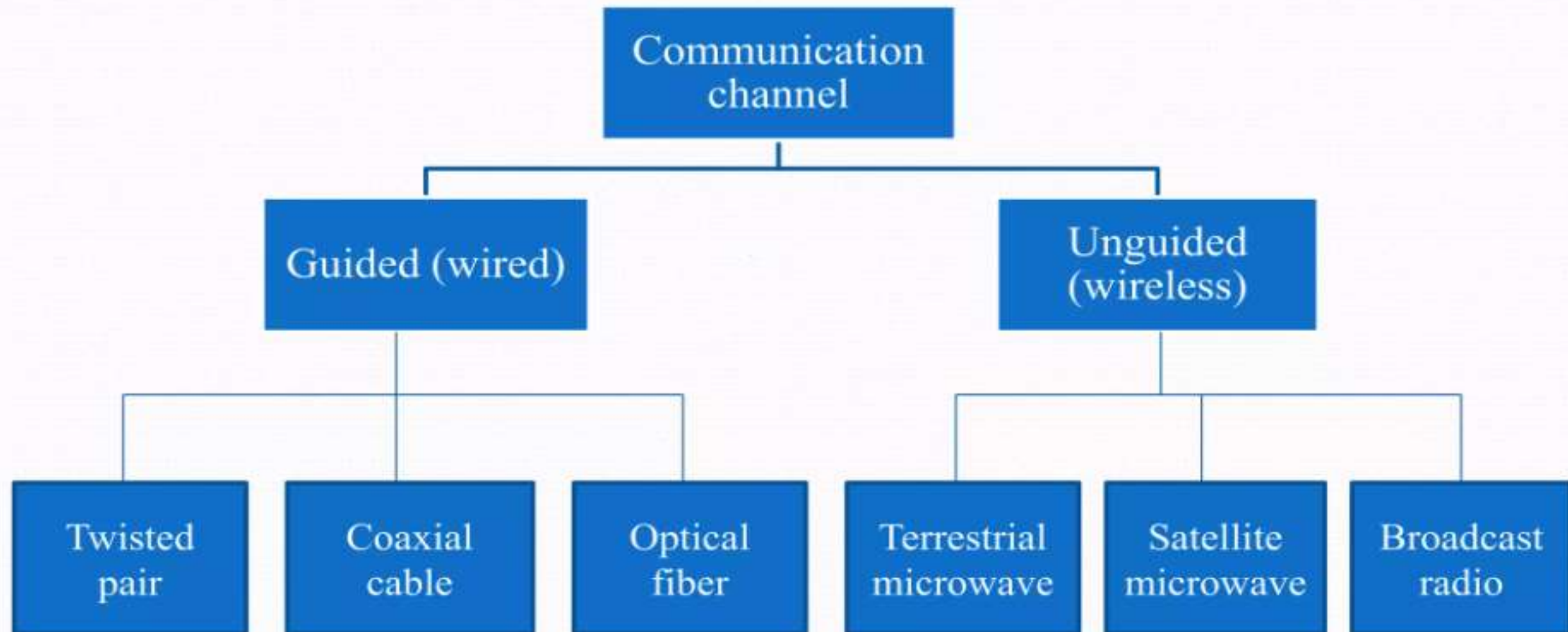
Multiple devices share the same link.

Lower installation cost.

Shared bandwidth.



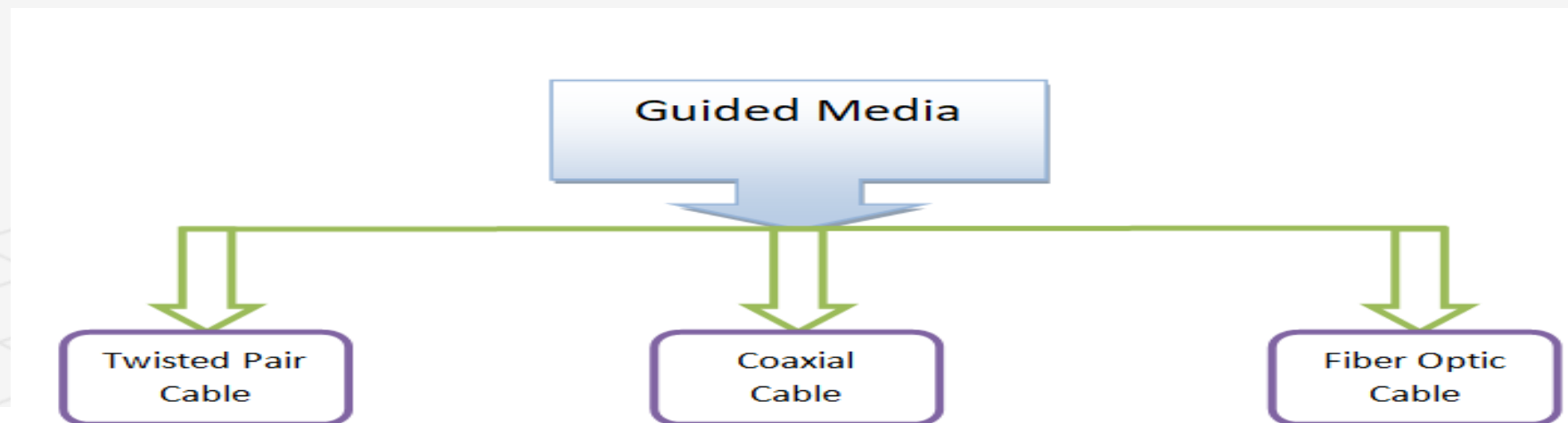
Types of Communication channel



Guided Media:

- In this transmission medium, the physical link is created using wires or cables between two or more computers or devices, and then the data is transmitted using these cables in terms of signals.
- It is defined as the physical medium through which the signals are transmitted.
- It is also known as Bounded media.

Guided media transmission of the following types:



Guided Media in Computer Networks

Sender

Receiver

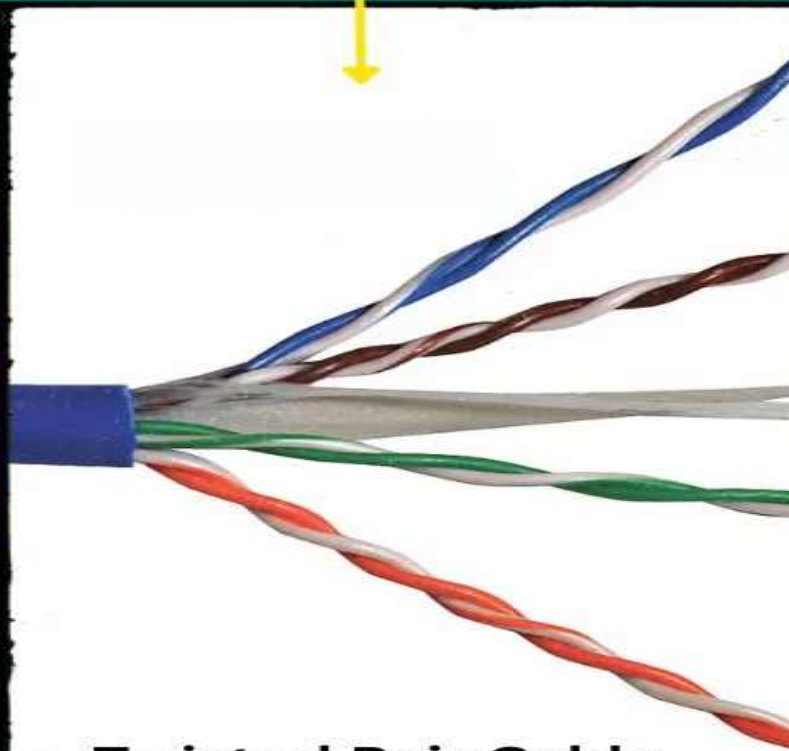
Physical Layer

Physical Layer

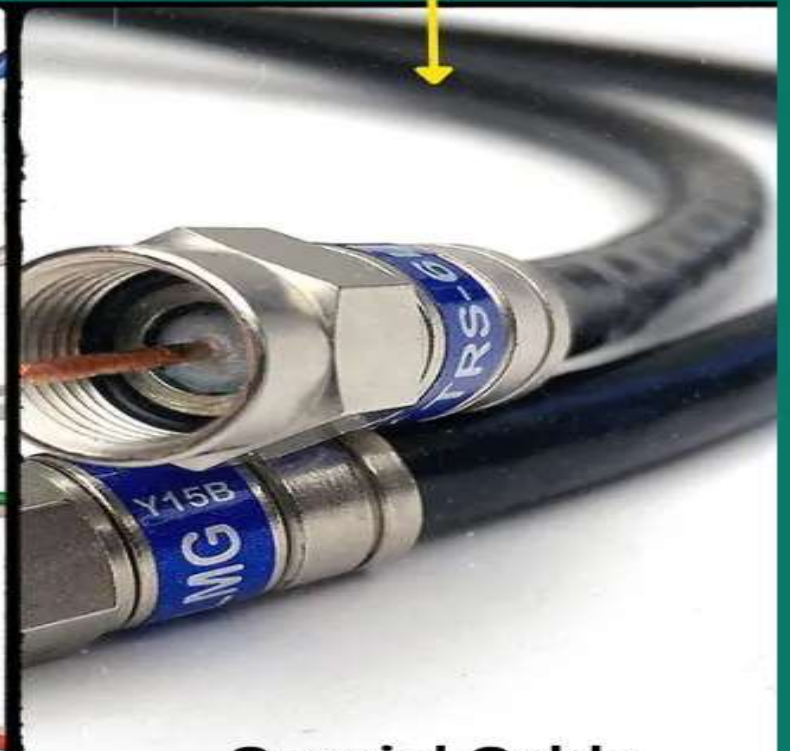
Guided Media



Fiber Optic Cable



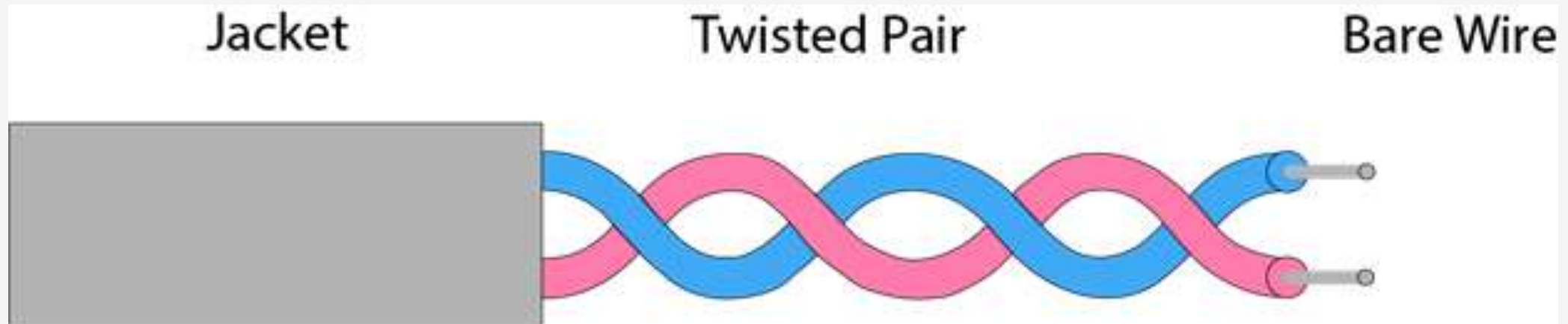
Twisted Pair Cable



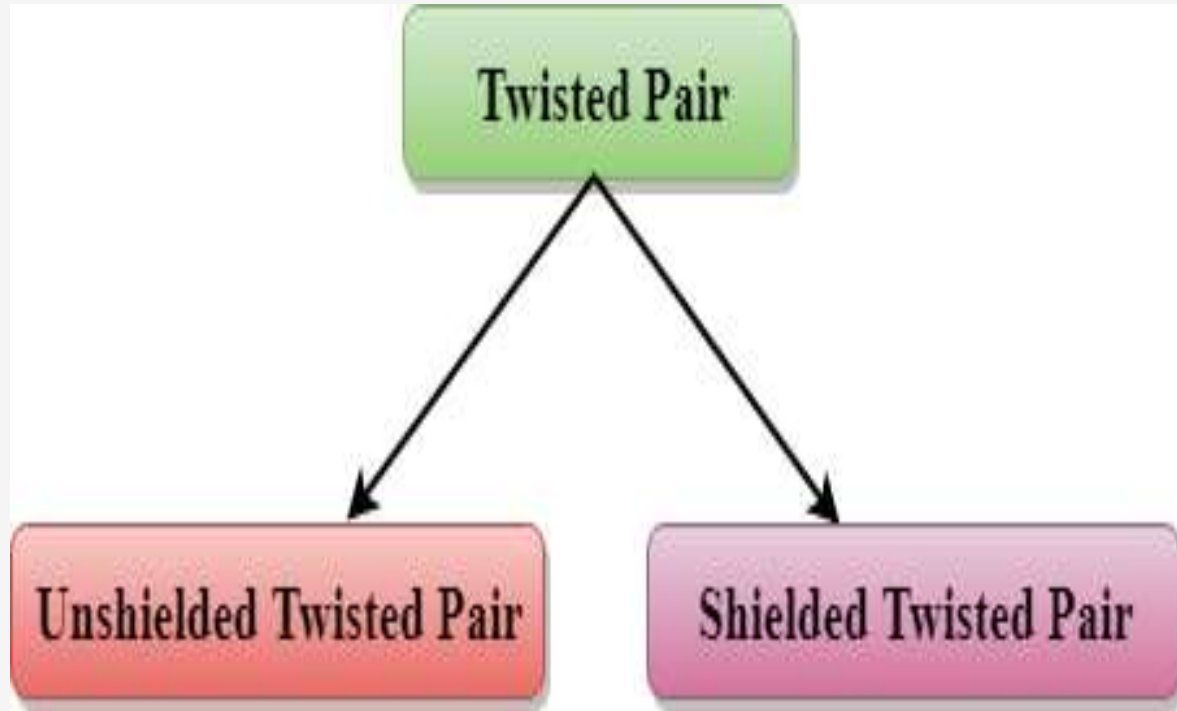
Coaxial Cable

Twisted pair:

- Twisted pair is a physical media made up of a pair of cables twisted with each other.
- A twisted pair cable is cheap as compared to other transmission media.
- Installation of the twisted pair cable is easy, and it is a lightweight cable.
- The frequency range for twisted pair cable is from 0 to 3.5KHz.
- A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern.
- The degree of reduction in noise interference is determined by the number of turns per foot.
- Increasing the number of turns per foot decreases noise interference.



Types of Twisted pair:



Unshielded Twisted Pair:

An unshielded twisted pair is widely used in telecommunication. Following are the categories of the unshielded twisted pair cable:

- ✓ Category 1: Category 1 is used for telephone lines that have low-speed data.
- ✓ Category 2: It can support upto 4Mbps.
- ✓ Category 3: It can support upto 16Mbps.
- ✓ Category 4: It can support upto 20Mbps.

Therefore, it can be used for long-distance communication.

- ✓ Category 5: It can support upto 200Mbps.



Unshielded Twisted Pair

Shielded Twisted Pair

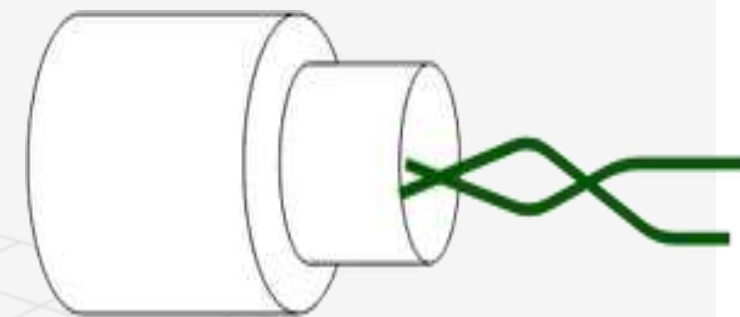
A shielded twisted pair is a cable that contains the mesh surrounding the wire that allows the higher transmission rate.

Characteristics Of Shielded Twisted Pair:

- The cost of the shielded twisted pair cable is not very high and not very low.
- An installation of STP is easy.
- It has higher capacity as compared to unshielded twisted pair cable.
- It has a higher attenuation.
- It is shielded that provides the higher data transmission rate.

Disadvantages

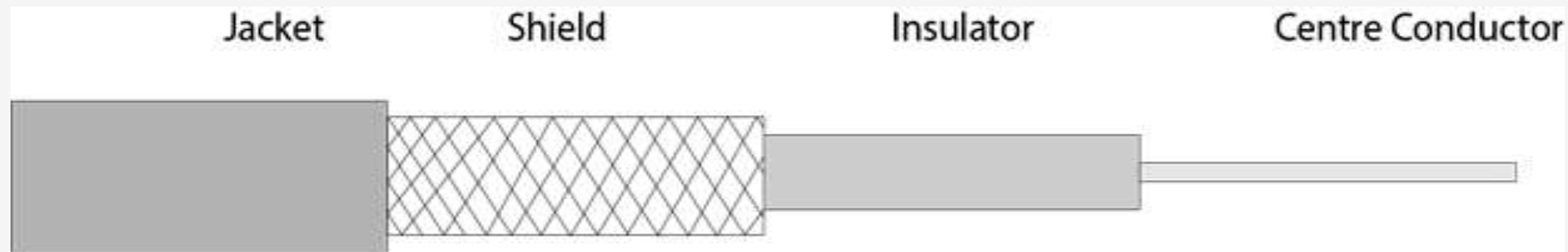
It is more expensive as compared to UTP and coaxial cable.
It has a higher attenuation rate.



Shielded Twisted Pair

Coaxial Cable

- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- The name of the cable is coaxial as it contains two conductors parallel to each other.
- It has a higher frequency as compared to Twisted pair cable.
- The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh.
- The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- The middle core is responsible for the data transferring whereas the copper mesh prevents from the EMI(Electromagnetic interference).

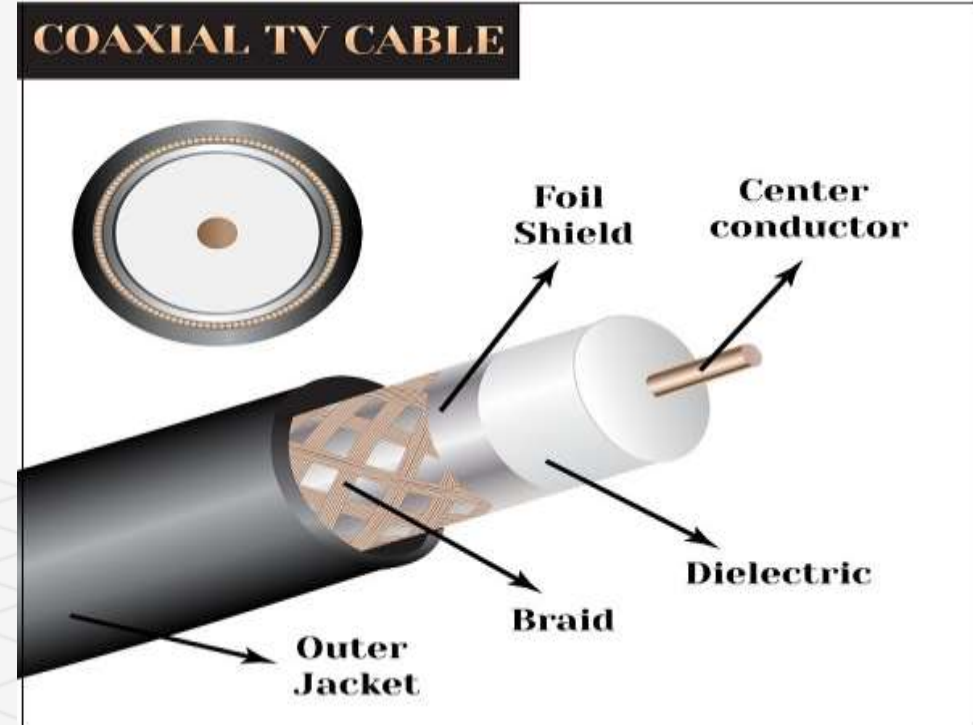


Advantages of Coaxial Cables

- ✓ Excellent noise immunity
- ✓ Signals can travel longer distances at higher speeds, e.g. 1 to 2 Gbps for 1 Km cable
- ✓ Can be used for both analog and digital signals
- ✓ Inexpensive as compared to fibre optic cables
- ✓ Easy to install and maintain

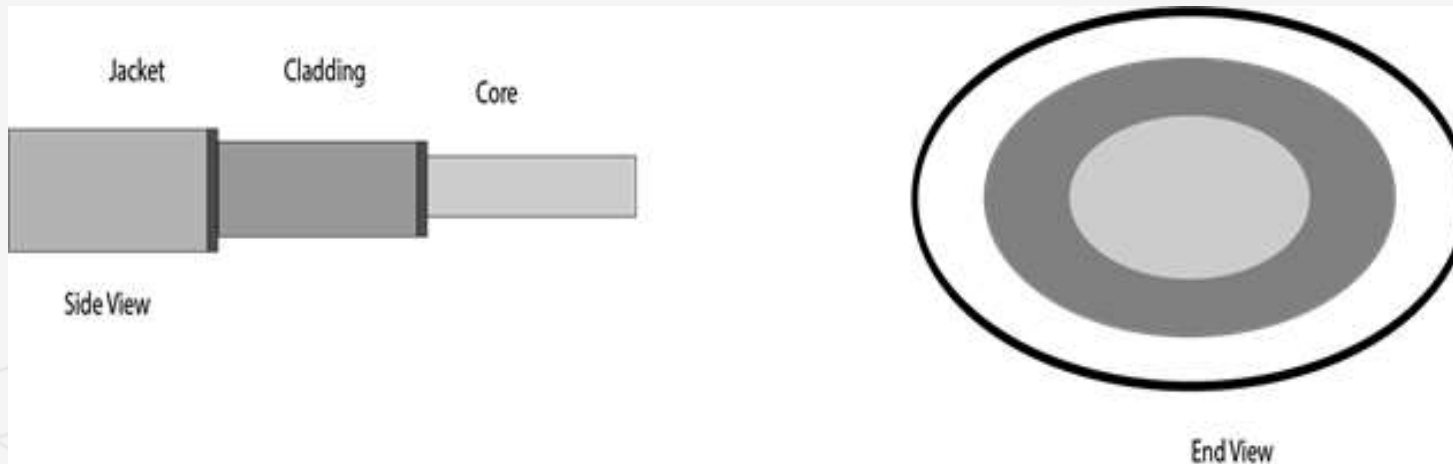
Disadvantages of Coaxial Cables

- ✓ Expensive as compared to twisted pair cables
- ✓ Not compatible with twisted pair cables



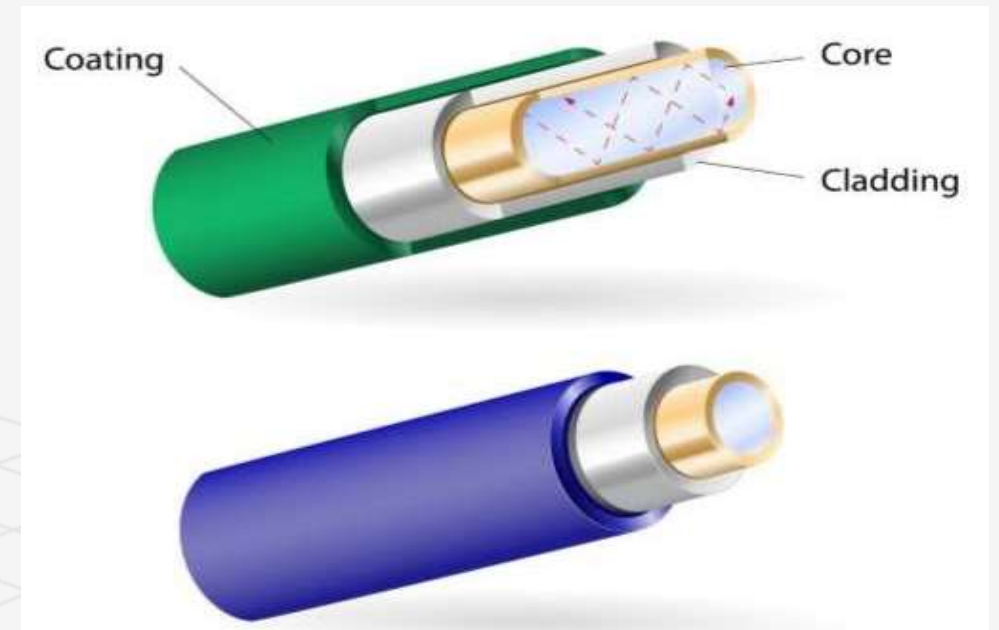
Fiber Optic

- ✓ Fiber optic cable is a cable that uses electrical signals for communication.
- ✓ Fiber optic is a cable that holds the optical fibres coated in plastic that are used to send the data by pulses of light.
- ✓ The plastic coating protects the optical fibers from heat, cold, electromagnetic interference from other types of wiring.
- ✓ Fiber optics provide faster data transmission than copper wires.



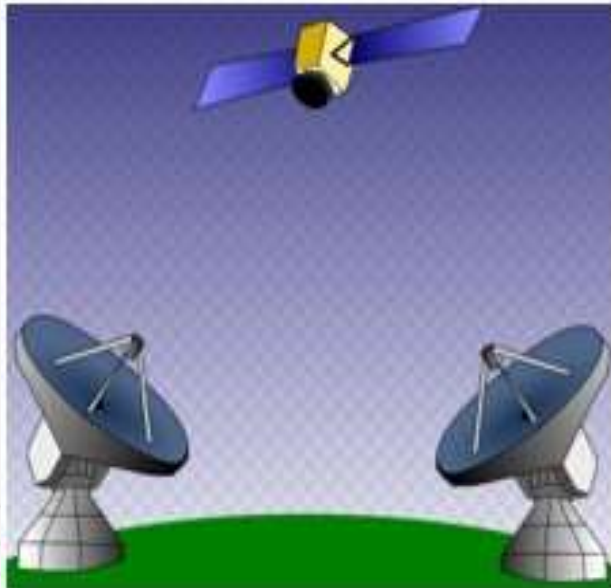
Fiber Optic

- **Core made** of high quality silica glass or plastic
- **Cladding made** of high quality silica glass or plastic, with a lower refractive index than the core
- Protective outer **covering** called buffer.



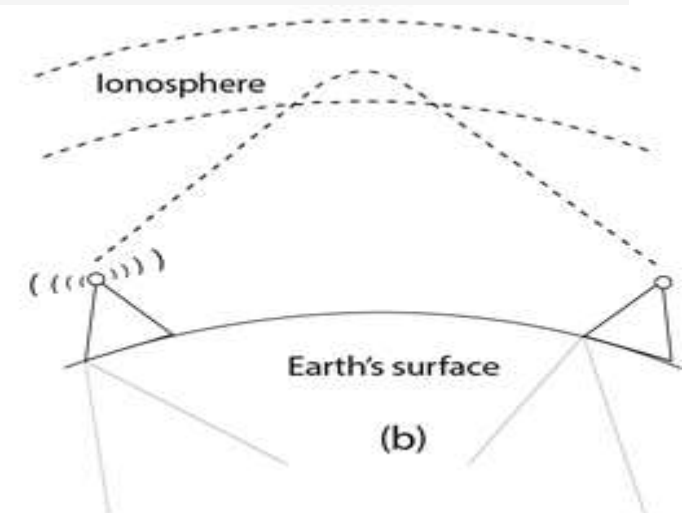
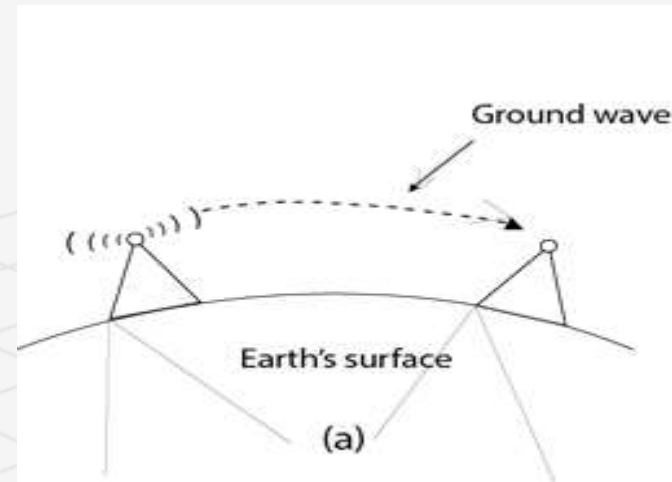
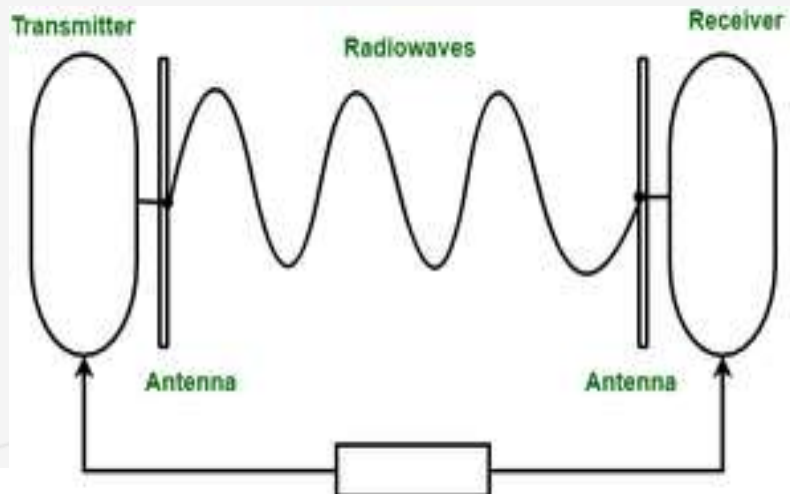
Unguided Transmission

- An unguided transmission transmits the electromagnetic waves without using any physical medium.
- Therefore it is also known as wireless transmission.
- The signal is broadcasted through air
- Less Secure
- Used for larger distances



Radio waves

- Radio waves are the electromagnetic waves that are transmitted in all the directions of free space.
- Radio waves are omnidirectional, i.e., the signals are propagated in all the directions.
- The range in frequencies of radio waves is from 3Khz to 1 khz.
- In the case of radio waves, the sending and receiving antenna are not aligned, i.e., the wave sent by the sending antenna can be received by any receiving antenna.
- An example of the radio wave is FM radio.



Applications Of Radio waves:

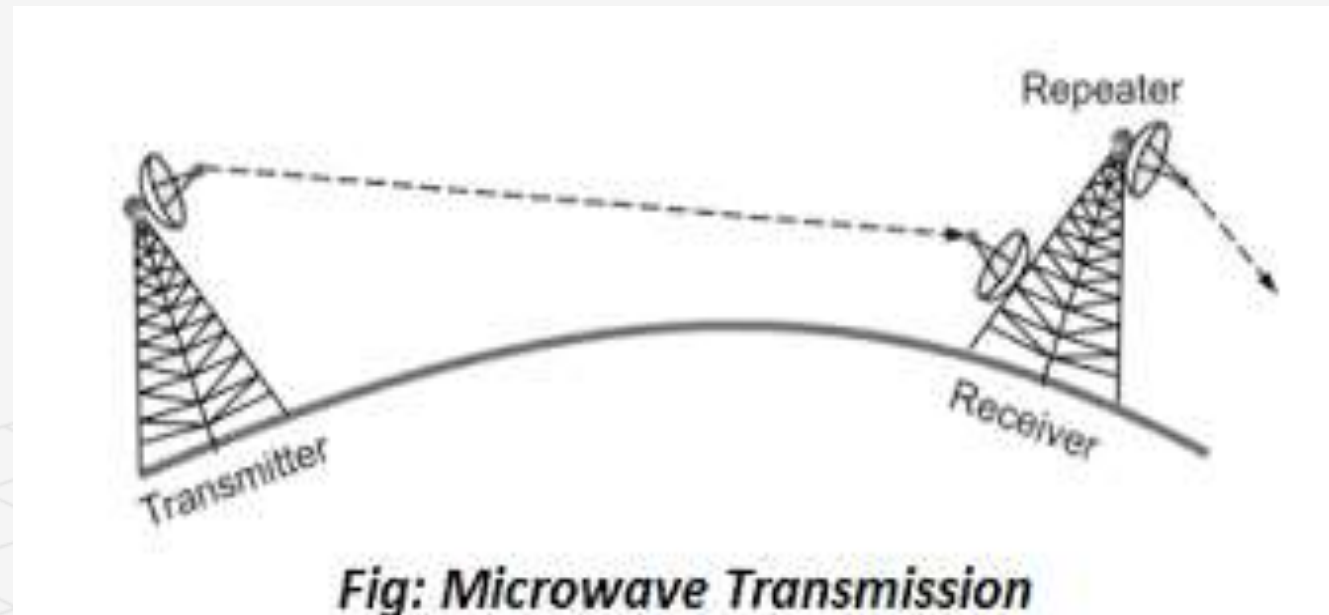
- A Radio wave is useful for multicasting when there is one sender and many receivers.
- An FM radio, television, cordless phones are examples of a radio wave.

Advantages Of Radio transmission:

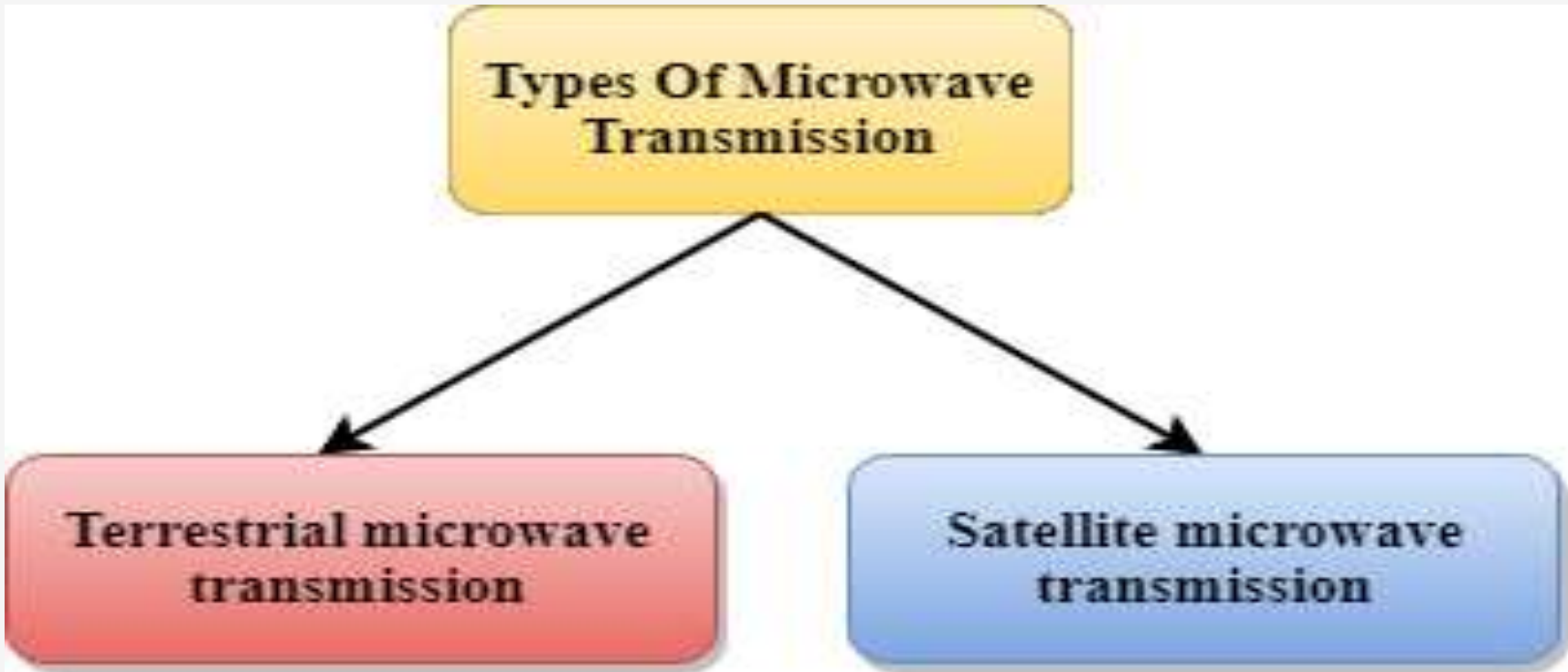
- Radio transmission is mainly used for wide area networks and mobile cellular phones.
 - Radio waves cover a large area, and they can penetrate the walls.
 - Radio transmission provides a higher transmission rate.
- 

Microwaves

- It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other.
- The distance covered by the signal is directly proportional to the height of the antenna.
- Frequency Range: 1GHz – 300GHz.
- These are majorly used for mobile phone communication and television distribution.



Types Of Microwave Transmission



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graph TD; A[Types Of Microwave Transmission] --> B[Terrestrial microwave transmission]; A --> C[Satellite microwave transmission];
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**Terrestrial microwave
transmission**

**Satellite microwave
transmission**

Terrestrial Microwave Transmission

- Terrestrial Microwave transmission is a technology that transmits the focused beam of a radio signal from one ground-based microwave transmission antenna to another.
 - Microwaves are the electromagnetic waves having the frequency in the range from 1GHz to 1000 GHz.
 - Microwaves are unidirectional as the sending and receiving antenna is to be aligned, i.e., the waves sent by the sending antenna are narrowly focused.
 - In this case, antennas are mounted on the towers to send a beam to another antenna which is km away.
 - It works on the line of sight transmission, i.e., the antennas mounted on the towers are the direct sight of each other.
- 

Satellite Microwave Communication

- A satellite is a physical object that revolves around the earth at a known height.
- Satellite communication is more reliable nowadays as it offers more flexibility than cable and fiber optic systems.
- We can communicate with any point on the globe by using satellite communication.

How Does Satellite work?

- The satellite accepts the signal that is transmitted from the earth station, and it amplifies the signal.
- The amplified signal is retransmitted to another earth station.

Infrared

- Infrared waves are used for very short distance communication.
- They cannot penetrate through obstacles.
- This prevents interference between systems.
- Frequency Range: 300GHz – 400THz.
- It is used in TV remotes, wireless mouse, keyboard, printer, etc.

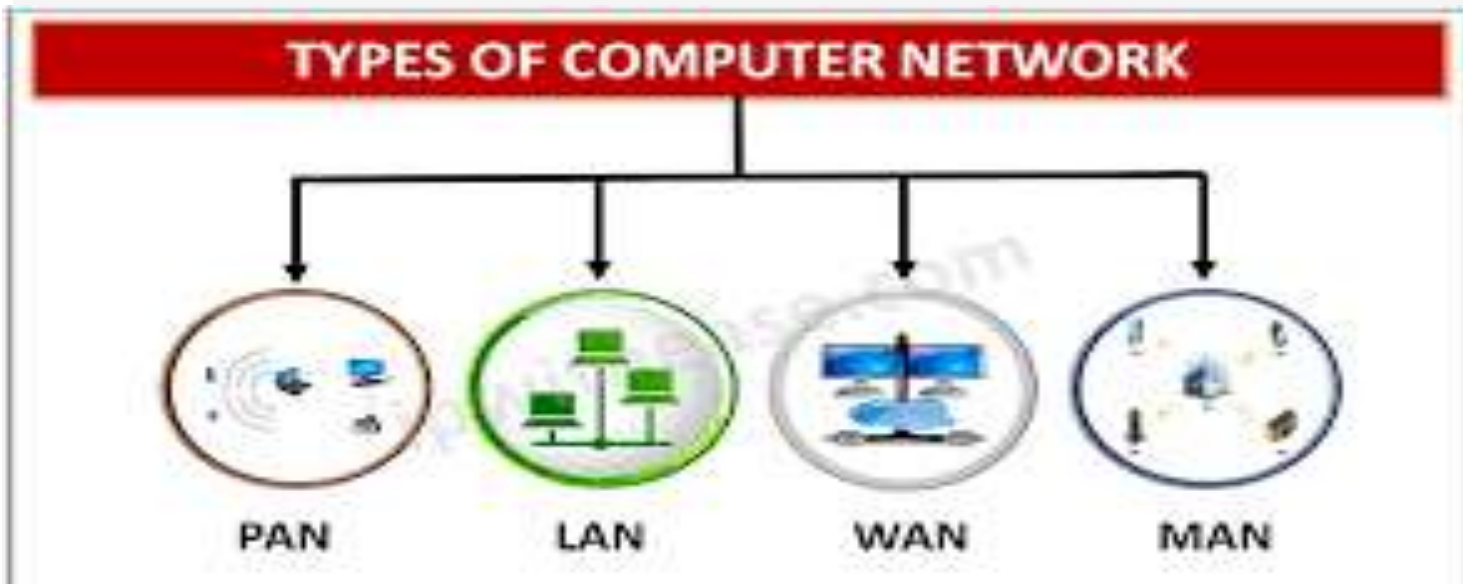


Twisted-Pair	Coaxial	Fiber-Optic	Infrared Light
			
Low Cost	Moderate Cost	High Cost	Moderate Cost
Best for short distances (330 ft.)	Moderate Distance (3300 ft. – thin) (8250 ft. – thick)	Long Distances (14,256 ft.)	Short distance (75 ft.)
Easy to Install	Professional Installation	Professional Installation	Easy to Install
Low Security	Average Security	High Security	Low Security
Low resistance to interference	Moderate resistance to interference	Very high resistance to interference	Very high resistance to interference

Computer Network Types - Categories of Networks

A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data, and applications.

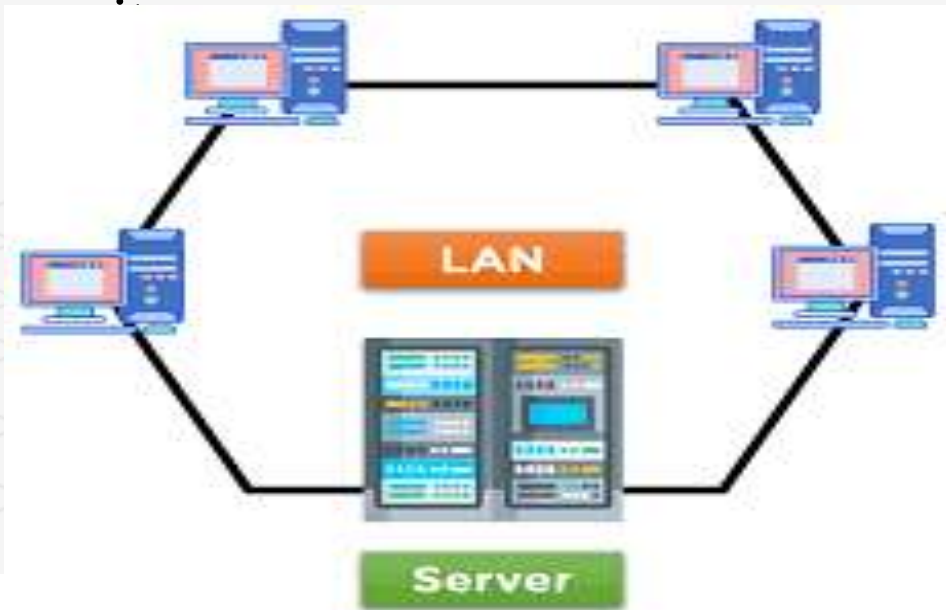
A computer network can be categorized by their size. A computer network is mainly of four types:



LAN(Local Area Network)
PAN(Personal Area Network)
MAN(Metropolitan Area Network)
WAN(Wide Area Network)

LAN(Local Area Network)

- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and ethernet cables.
- The data is transferred at an extremely faster rate in Local Area Network.
- Local Area Network provides higher s



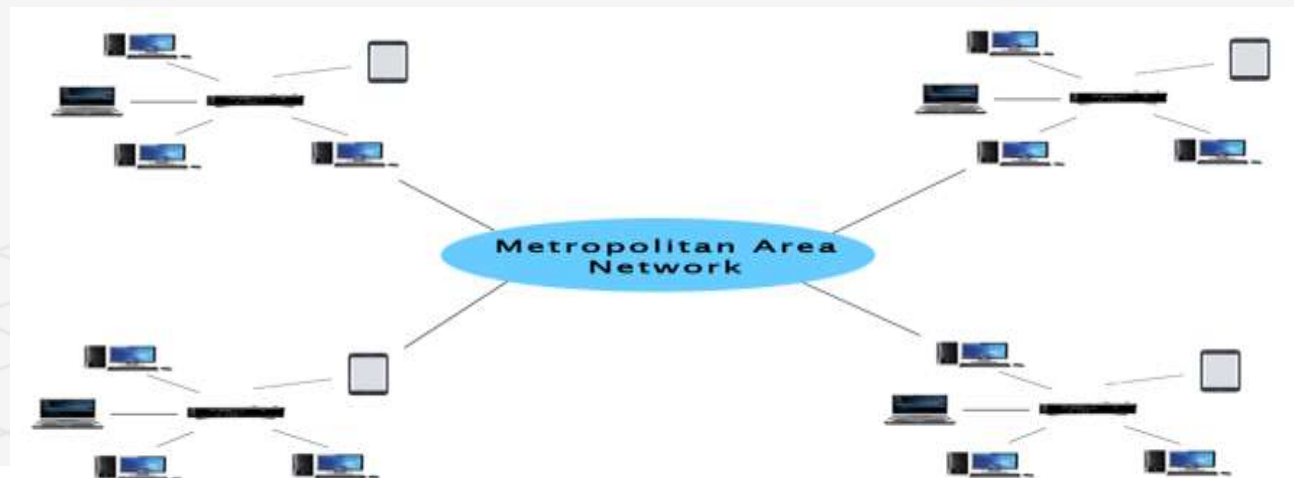
PAN(Personal Area Network)

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- Thomas Zimmerman was the first research scientist to bring the idea of the Personal Area Network.
- Personal Area Network covers an area of 30 feet.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.



MAN(Metropolitan Area Network)

- A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- Government agencies use MAN to connect to the citizens and private industries.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- The most widely used protocols in MAN are RS-232, Frame Relay, ATM, ISDN, OC-3, ADSL, etc.
- It has a higher range than Local Area Network(LAN).

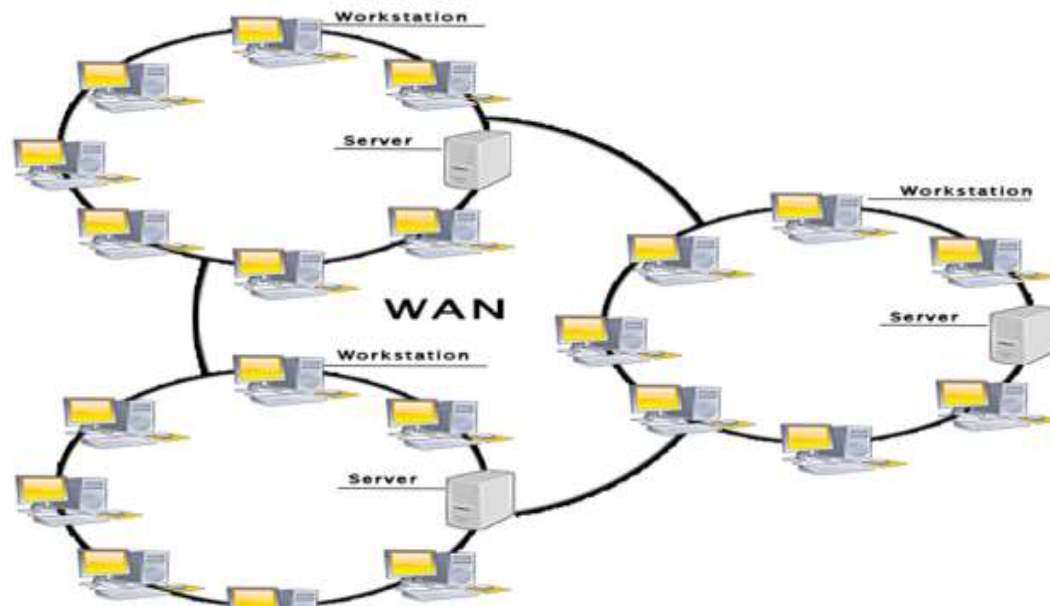


Uses Of Metropolitan Area Network:

- MAN is used in communication between the banks in a city.
- It can be used in an Airline Reservation.
- It can be used in a college within a city.
- It can also be used for communication in the military.

WAN(Wide Area Network)

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fiber optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.



Advantages Of Wide Area Network:

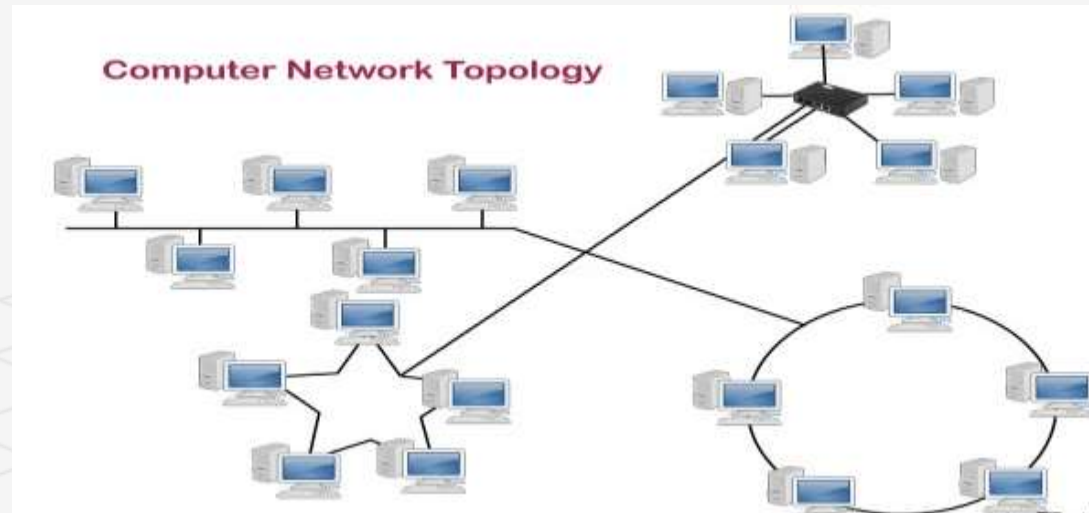
- **Geographical area:** A Wide Area Network provides a large geographical area. Suppose if the branch of our office is in a different city then we can connect with them through WAN. The internet provides a leased line through which we can connect with another branch.
- **Centralized data:** In case of WAN network, data is centralized. Therefore, we do not need to buy the emails, files or back up servers.
- **Get updated files:** Software companies work on the live server. Therefore, the programmers get the updated files within seconds.
- **Exchange messages:** In a WAN network, messages are transmitted fast. The web application like Facebook, Whatsapp, Skype allows you to communicate with friends.
- **Sharing of software and resources:** In WAN network, we can share the software and other resources like a hard drive, RAM.
- **Global business:** We can do the business over the internet globally.
- **High bandwidth:** If we use the leased lines for our company then this gives the high bandwidth. The high bandwidth increases the data transfer rate which in turn increases the productivity of our company.

Disadvantages of Wide Area Network:

- **Security issue:** A WAN network has more security issues as compared to LAN and MAN network as all the technologies are combined together that creates the security problem.
- **Needs Firewall & antivirus software:** The data is transferred on the internet which can be changed or hacked by the hackers, so the firewall needs to be used. Some people can inject the virus in our system so antivirus is needed to protect from such a virus.
- **High Setup cost:** An installation cost of the WAN network is high as it involves the purchasing of routers, switches.
- **Troubleshooting problems:** It covers a large area so fixing the problem is difficult.

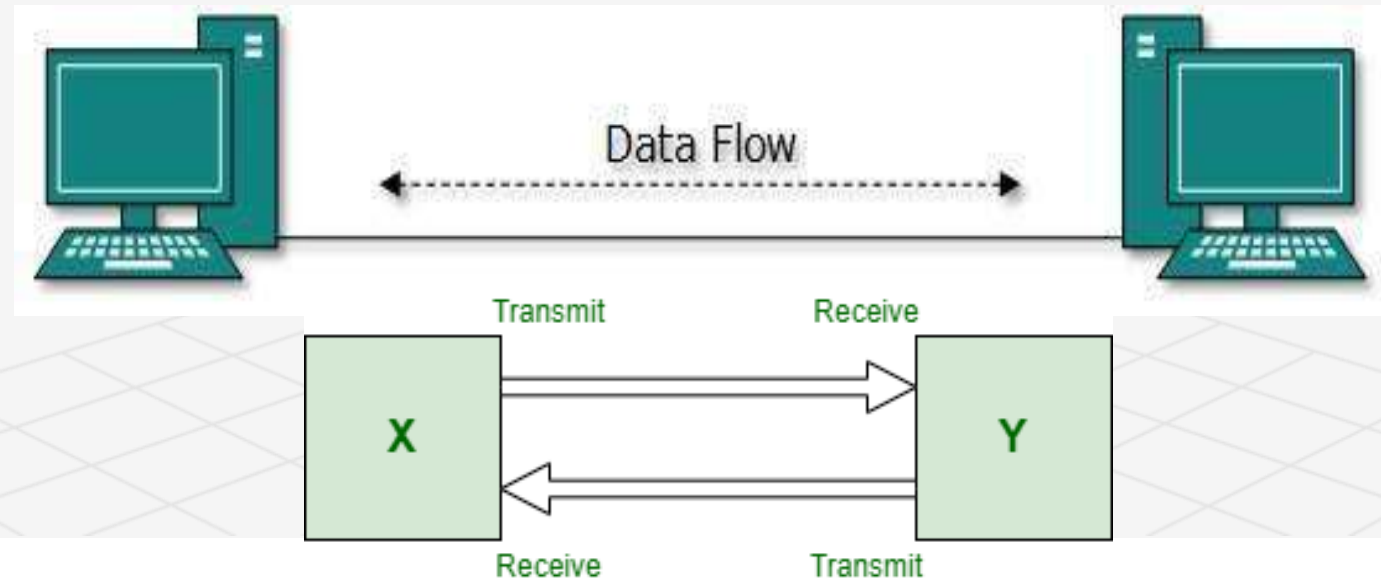
What is Network Topology? – Physical Structures

- A Network Topology is the arrangement with which computer systems or network devices are connected to each other.
- Topologies may define both physical and logical aspect of the network.
- Both logical and physical topologies could be same or different in a same network.



Point to Point Topology

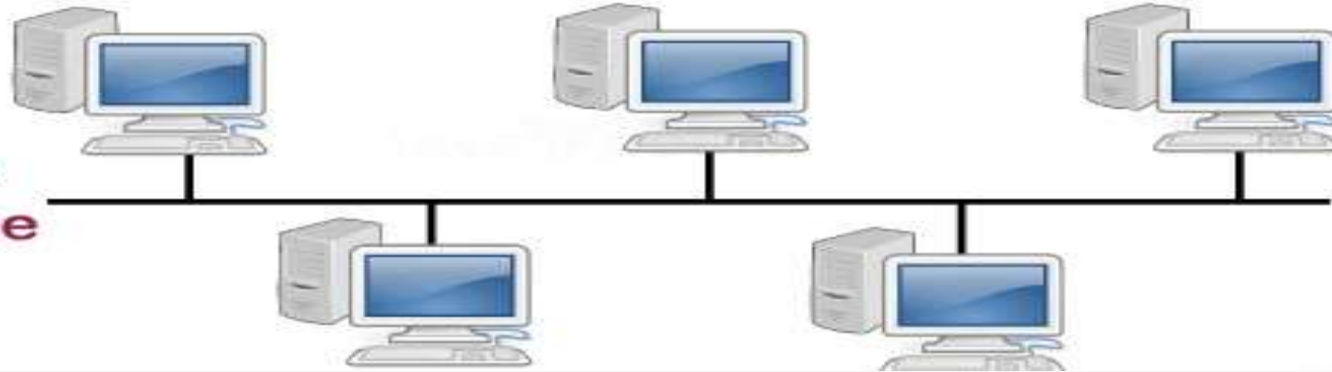
- Point-to-Point Topology is a type of topology that works on the functionality of the sender and receiver.
- It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver.
- Point-to-Point provides high bandwidth.
- Point-to-point networks contains exactly two hosts such as computer, switches or routers, servers connected back to back using a single piece of cable.



Bus Topology

Network backbone

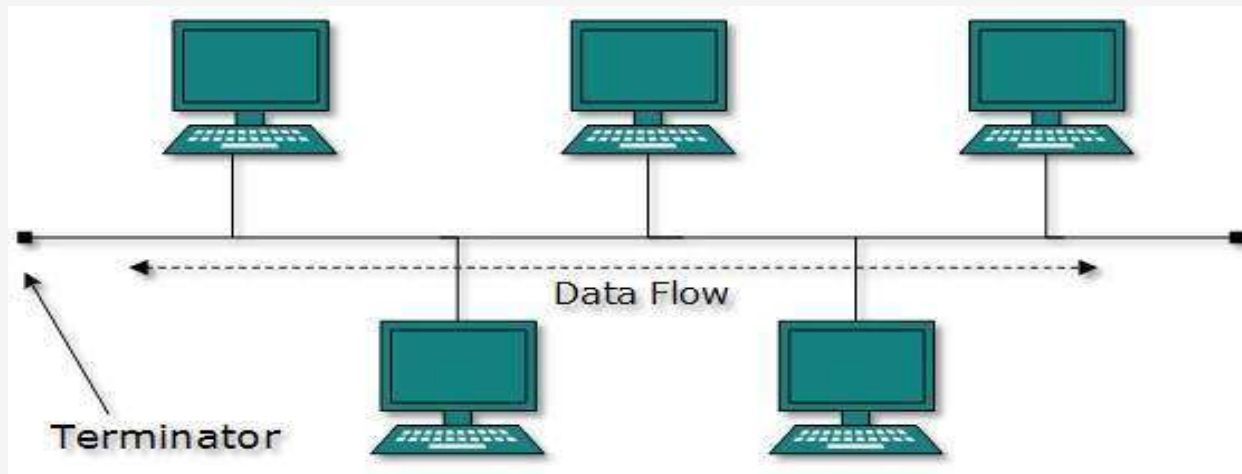
BUS



- The bus topology is designed in such a way that all the stations are connected through a single cable known as a backbone cable.
- Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.
- When a node wants to send a message over the network, it puts a message over the network.
- All the stations available in the network will receive the message whether it has been addressed or not.

Bus Topology

- The bus topology is mainly used in 802.3 (ethernet) and 802.4 standard networks.
- The configuration of a bus topology is quite simpler as compared to other topologies.
- The backbone cable is considered as a "single lane" through which the message is broadcast to all the stations.
- The most common access method of the bus topologies is CSMA (Carrier Sense Multiple Access).



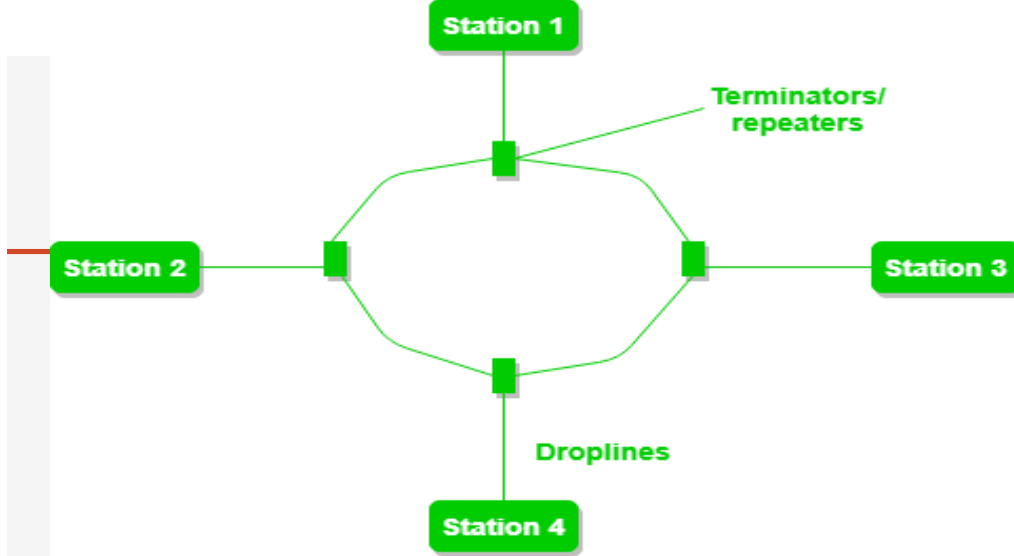
Advantages of Bus Topology :

- It is the easiest network topology for connecting peripherals or computers in a linear fashion.
- It works very efficiently well when there is a small network.
- The length of cable required is less than a star topology.
- It is easy to connect or remove devices in this network without affecting any other device.
- Very cost-effective as compared to other network topology i.e. mesh and star
- It is easy to understand topology.
- Easy to expand by joining the two cables together.

Disadvantages of Bus Topology :

- Bus topology is not great for large networks.
- Identification of problems becomes difficult if the whole network goes down.
- Troubleshooting individual device issues is very hard.
- Need terminators are required at both ends of the main cable.
- Additional devices slow the network down.
- If the main cable is damaged, the whole network fails or splits into two.
- Packet loss is high.
- This network topology is very slow as compared to other topologies.

Ring Topology



- Ring topology is like a bus topology, but with connected ends.
- The node that receives the message from the previous computer will retransmit to the next node.
- The data flows in one direction, i.e., it is unidirectional.
- The data flows in a single loop continuously known as an endless loop.
- It has no terminated ends, i.e., each node is connected to other node and having no termination point.
- The data in a ring topology flow in a clockwise direction.

➤ Ring Topology

- The most common access method of the ring topology is token passing.
- Token passing: It is a network access method in which token is passed from one node to another node.
- Token: It is a frame that circulates around the network.

Working of Token passing

- A token moves around the network, and it is passed from computer to computer until it reaches the destination.
- The sender modifies the token by putting the address along with the data.
- The data is passed from one device to another device until the destination address matches.
- Once the token received by the destination device, then it sends the acknowledgment to the sender.
- In a ring topology, a token is used as a carrier.

Star Topology

Star Topology

- All hosts in Star topology are connected to a central device, known as hub device, using a point-to-point connection.
- Point-to-point connection between hosts and hub.



- Star topology is an arrangement of the network in which every node is connected to the central hub, switch or a central computer.
- The central computer is known as a server, and the peripheral devices attached to the server are known as clients.
- This hub is the central node and all other nodes are connected to the central node.
- Coaxial cable or RJ-45 cables are used to connect the computers.
- Hubs or Switches are mainly used as connection devices in a physical star topology.
- Star topology is the most popular topology in network implementation.

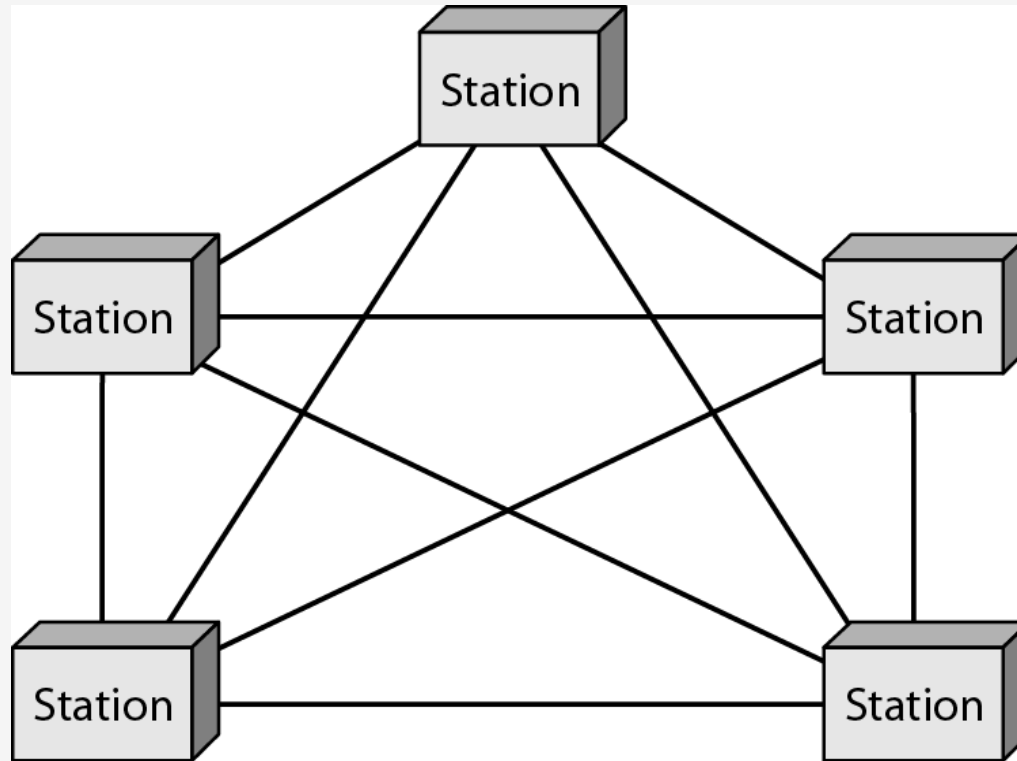
Advantages of Star topology

- **Efficient troubleshooting:** Troubleshooting is quite efficient in a star topology as compared to bus topology. In a bus topology, the manager has to inspect the kilometers of cable. In a star topology, all the stations are connected to the centralized network. Therefore, the network administrator has to go to the single station to troubleshoot the problem.
- **Network control:** Complex network control features can be easily implemented in the star topology. Any changes made in the star topology are automatically accommodated.
- **Limited failure:** As each station is connected to the central hub with its own cable, therefore failure in one cable will not affect the entire network.
- **Familiar technology:** Star topology is a familiar technology as its tools are cost-effective.
- **Easily expandable:** It is easily expandable as new stations can be added to the open ports on the hub.
- **Cost effective:** Star topology networks are cost-effective as it uses inexpensive coaxial cable.
- **High data speeds:** It supports a bandwidth of approx 100Mbps. Ethernet 100BaseT is one of the most popular Star topology networks.

Disadvantages of Star topology

- **A Central point of failure:** If the central hub or switch goes down, then all the connected nodes will not be able to communicate with each other.
- **Cable:** Sometimes cable routing becomes difficult when a significant amount of routing is required.

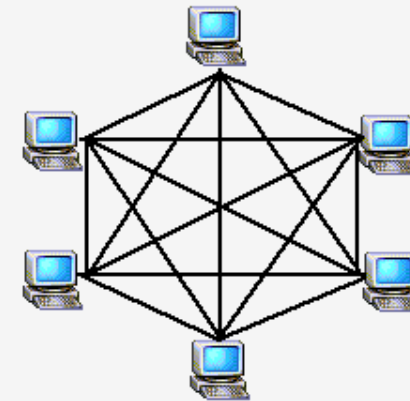
A fully connected mesh topology (five devices)



- *Every link is dedicated point-to-point link*
- *The term dedicated means that the link carries traffic only between the two devices it connects*
- *To accommodate that many links, every device must have $n-1$ input/output ports*

✓ To link n devices fully connected mesh has:
 $n(n-1)/2$ physical channels (Full-Duplex)

✓ Every Device on the network must have $n-1$ ports



Mesh topology (Eight devices)

■ Example:

8 devices in mesh has links: $n(n-1) / 2$

number of links = $8(8-1)/2 = 28$

number of ports per device = $n - 1 = 8 - 1 = 7$

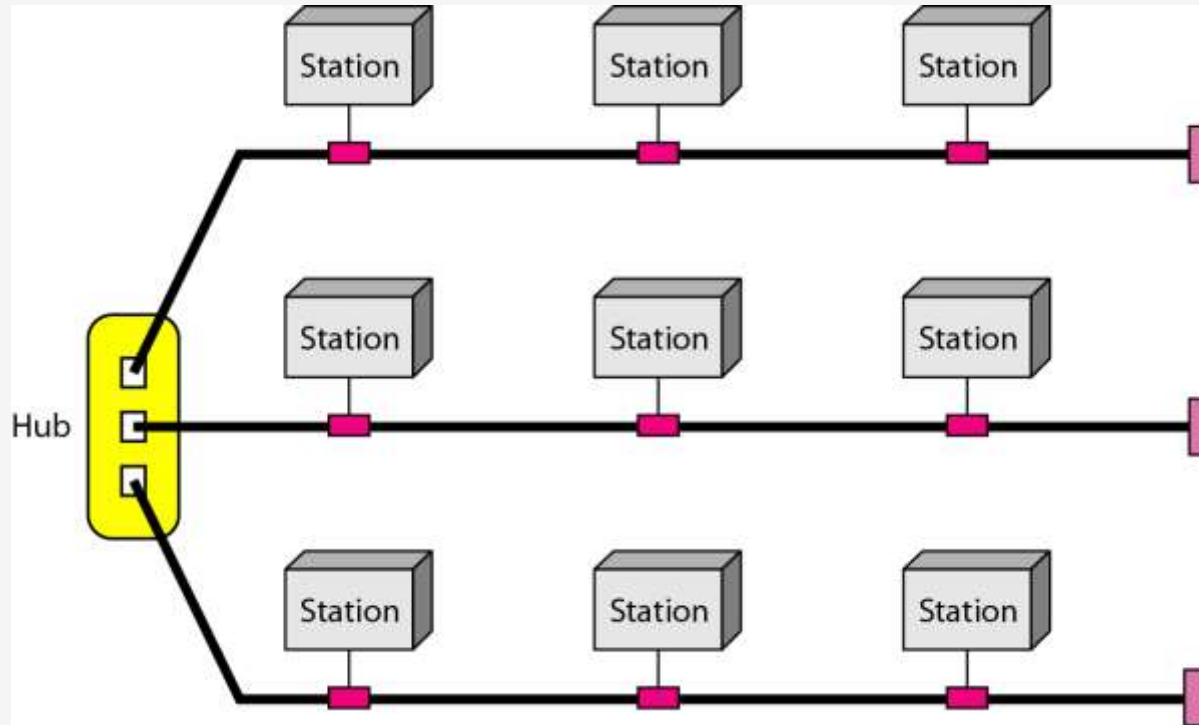
Advantages

- *Each connection carry its own data load (no traffic problems)*
- *A mesh topology is robust*
- *Privacy or security*
- *Fault identification and fault isolation*

Disadvantages

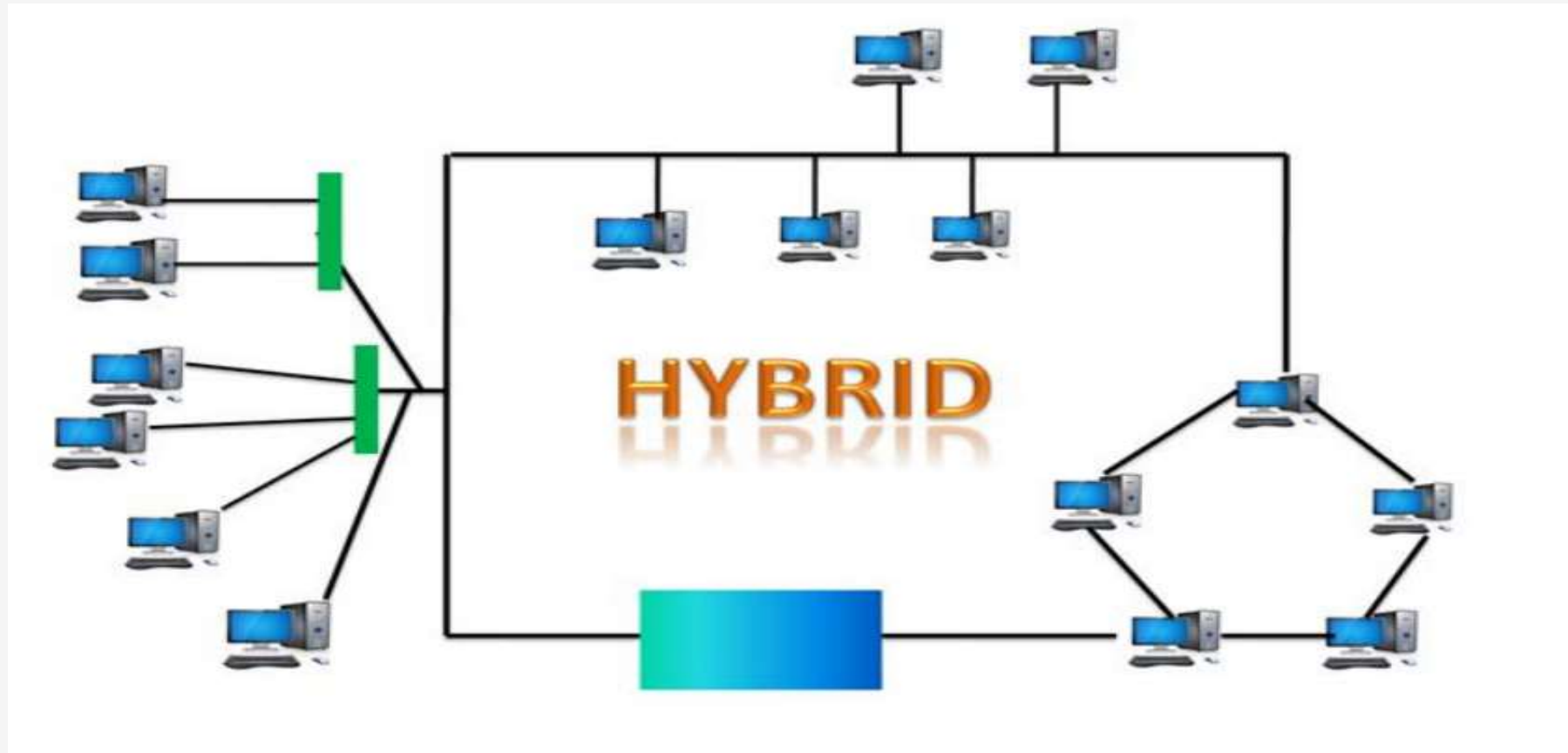
- *Big amount of cabling*
- *Big number of I/O ports*
- *Installation and reconnection are difficult*
- *Sheer bulk of the wiring can be greater than the available space*
- *Hardware connect to each I/O could be expensive*

A hybrid topology: a star backbone with three bus networks



Example: *having a main star topology with each branch connecting several stations in a bus topology*

Hybrid Topology



Exercises

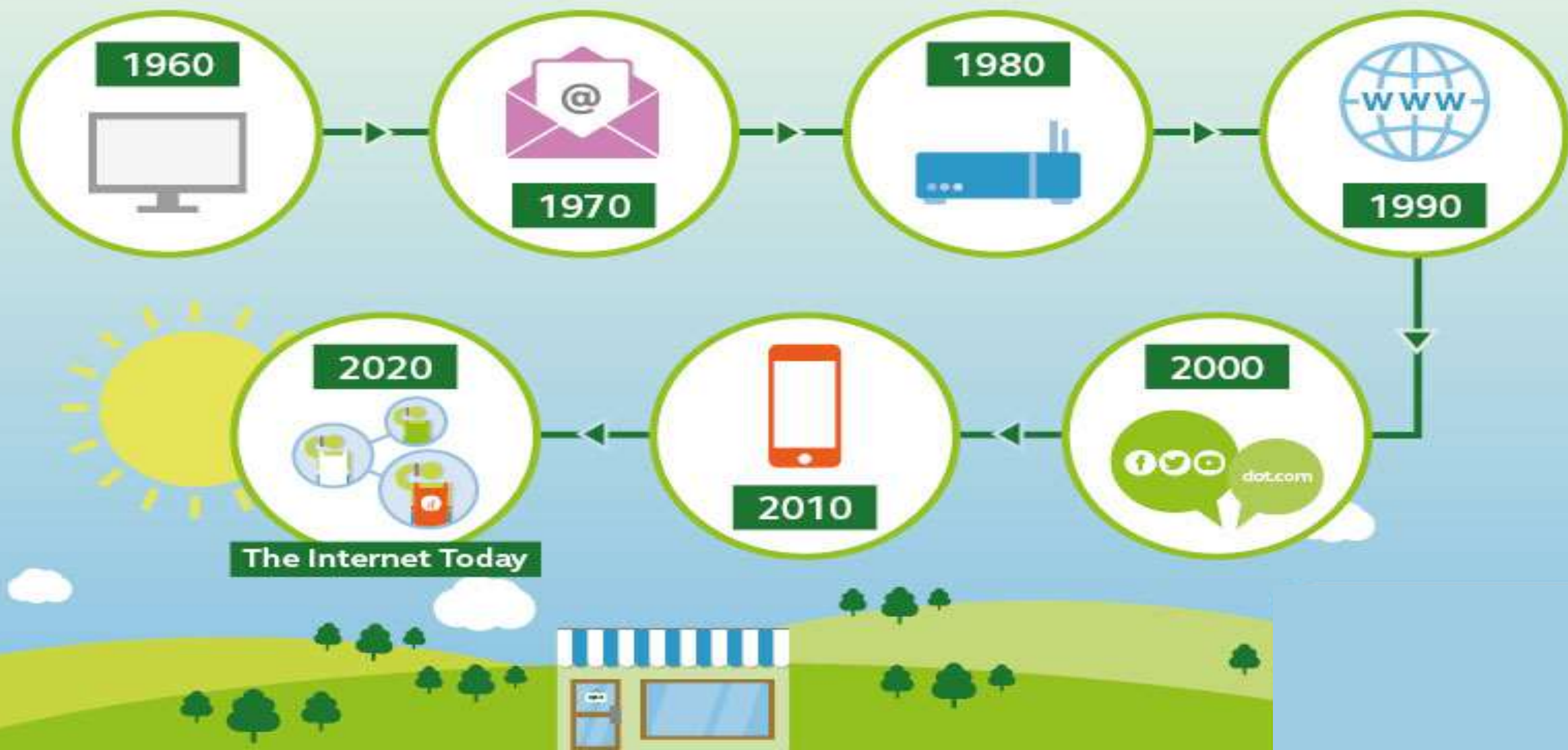
1. A Full Mesh network has 8 computers. Find the number of cable links required.
2. A university network has one central switch connected to all classrooms.
 - a) Which topology is used?
 - b) What happens if one classroom cable is damaged?
 - c) What happens if the switch fails?
3. An organization wants maximum reliability because the network controls hospital equipment. Which topology should be selected?
4. A company wants to minimize installation cost for 10 computers. Which topology should they choose?
5. A school plans to connect **30 computers**.
Requirements:
Easy maintenance
Easy expansion
Good performance
Which topology would you recommend?

-
6. How many ports are required on each device in a Mesh topology with 6 computers?
Each device connects to every other device.
 7. A Mesh network has 45 cable links. How many computers are connected?

What is Internet?

- The Internet is the foremost important tool and the prominent resource that is being used by almost every person across the globe.
- It connects millions of computers, webpages, websites, and servers.
- Using the internet we can send emails, photos, videos, and messages to our loved ones.
- Internet is a widespread interconnected network of computers and electronic devices(that support Internet).
- It creates a communication medium to share and get information online.
- If your device is connected to the Internet then only you will be able to access all the applications, websites, social media apps, and many more services.
- The Internet nowadays is considered the fastest medium for sending and receiving information.

History of the Internet



Protocols

- A network protocol is an accepted set of rules that govern data communication between different devices in the network.
- It determines what is being communicated, how it is being communicated, and when it is being communicated.
- It permits connected devices to communicate with each other, irrespective of internal and structural differences.
- Network protocols are like a common language for computers.
- The computers within a network may use vastly different software and hardware;
- however, the use of protocols enables them to communicate with each other regardless.

Key Elements of Protocol

Syntax : syntax refers to the structure or the format of the data that gets exchanged between the devices. Syntax of message includes the type of data, composition of message and sequencing of message. The starting 8 bits of data is considered as the address of the sender. The next 8 bits is considered to be the address of the receiver. The remaining bits are considered as the message itself.

Semantics : Semantics defines data transmitted between devices. It provides rules and norms for understanding message or data element values and actions.

Timing : Timing refers to the synchronization and coordination between devices while transferring the data. Timing ensures at what time data should be sent and how fast data can be sent. For example, If a sender sends 100 Mbps but the receiver can only handle 1 Mbps, the receiver will overflow and lose data. Timing ensures preventing data loss, collisions and other timing related issues.

Key Elements of Protocol

Sequence control : Sequence control ensures the proper ordering of data packets. The main responsibility of sequence control is to acknowledge the data while it get received, and the retransmission of lost data. Through this mechanism the data is delivered in correct order.

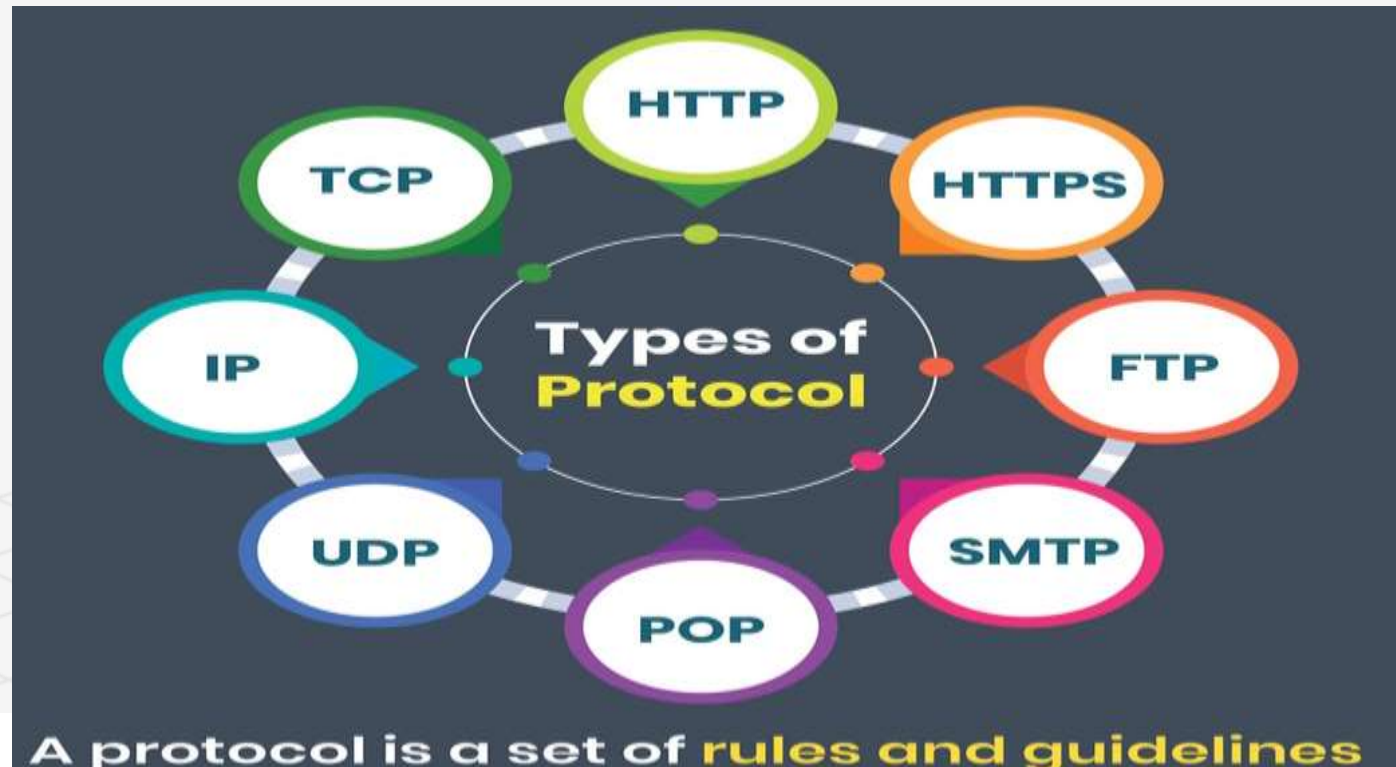
Flow Control : Flow control regulates device data delivery. It limits the sender's data or asks the receiver if it's ready for more. Flow control prevents data congestion and loss.

Error Control : Error control mechanisms detect and fix data transmission faults. They include error detection codes, data resend, and error recovery. Error control detects and corrects noise, interference, and other problems to maintain data integrity.

Security : Network security safeguards data confidentiality, integrity, and authenticity. which includes encryption, authentication, access control, and other security procedures. Network communication's privacy and trustworthiness are protected by security standards.

Types of network protocols

- Different protocols serve different functions to ensure efficient, quick, and secure network communication.
- Various types of network protocols can be categorized into the following three broad categories to help organizations operate seamlessly across different business scenarios:



Network Communication Protocols:

- These protocols determine the rules and formats to transfer data across networks.
- Communication protocols govern various aspects of analog and digital communications, such as syntax, authentication, semantics, and error detection, among others.

Some key network communication protocols include:

- **Hyper-Text Transfer Protocol (HTTP):** Commonly referred to as the protocol of the internet that allows communication between a server and browser.
- **Transmission Control Protocol (TCP):** A reliable, connection-oriented protocol that helps in the sequential transmission of data packets to ensure data reaches the destination on time without duplication.
- **Internet Protocol (IP):** Facilitates routing the data packets across networks. IP contains addressing and control information to deliver packets across a network. It works along with TCP. While it ensures delivering the packets to the right address, TCP aligns them in the right order.

-
- **User Datagram Protocol (UDP):** Unlike TCP, UDP is a connectionless protocol that doesn't ensure a connection between the application and server before transmitting a message. It's effective for use cases such as broadcasts or multicast connections.
 - **File Transfer Protocol (FTP):** Allows file sharing between servers by establishing two TCP connections, one for data transfer and the other for control. The data transfer connection transfers the actual files while the control connection transfers control information such as passwords to ensure data retrieval in case of data loss.
- 

Network Security Protocols:

- These protocols ensure safe data transmission over the network connections.
- Network security protocols define the procedures to secure data from any unauthorized access.
- These protocols leverage encryption and cryptography to safeguard.

Here are the most widely used network security protocols:

- **Secure File Transfer Protocol (SFTP):** Helps securely transfer files across a network by using public-key encryption and authenticating the client and server.
- **Hyper-Text Transfer Protocol Secure (HTTPS):** Overcomes the limitation of HTTP by ensuring the security of data transmitted between the browser and server through data encryption. HTTPS is a secure version of HTTP.
- **Secure Socket Layer (SSL):** Primarily helps secure internet connections and safeguard sensitive data using encryption. SSL protocol enables both server-client communication and server-server communication.

Network Management Protocols:

- Network managers require standard policies and procedures to manage and monitor the network for maintaining smooth communication.
- Network management protocols ensure quick troubleshooting and optimal performance across the network.

The following are essential network protocols management:

- **Simple Network Management Protocol (SNMP):** Helps administrators manage network devices by monitoring endpoint information to proactively track network performance and pinpoint network glitches for quick troubleshooting.
- **Internet Control Message Protocol (ICMP):** Helps diagnose network connectivity issues. Network devices employ ICMP for sending error messages, highlighting congestion and timeouts, and transmitting other operational information to assist in network troubleshooting.

How do network protocols work?

- Network protocols split the communication process into discrete tasks across each OSI model layer.
 - To enable network communication, one or more protocols operate at every layer.
 - For example, the Internet Protocol (IP) routes data by managing the information such as data packets' source address and destination to enable network-to-network communications. Therefore, it's referred to as a network layer protocol.
- 

Standards

Standards are the set of rules for data communication that are needed for exchange of information among devices. It is important to follow Standards which are created by various Standard Organization like IEEE , ISO , ANSI etc.

Types of Standards

De Facto Standard.

De Jure Standard.

De Facto Standard : The meaning of the word ” De Facto ” is ” By Fact ” or “By Convention”. These are the standards that have not been approved by any Organization, but have been adopted as Standards because of it’s widespread use. Also , sometimes these standards are often established by Manufacturers.

For example : Apple and Google are two companies which established their own rules on their products which are different . Also they use some same standard rules for manufacturing for their products.

De Jure Standard : The meaning of the word “De Jure” is “By Law” or “By Regulations”. Thus , these are the standards that have been approved by officially recognized body like ANSI , ISO , IEEE etc. These are the standard which are important to follow if it is required or needed.

For example : All the data communication standard protocols like SMTP , TCP , IP , UDP etc. are important to follow the same when we needed them.

Network Standards Organizations

ANSI	American National Standards Institute
IEEE	Institute of Electrical and Electronics Engineers
ITU	International Telecommunications Union
ISO	International Organization for Standardization
ISOC	Internet Society
IETF	Internet Society and the Internet Engineering Task Force
EIA/TIA	Electronic Industries Alliance and the Telecommunications Industry Association

Networking hardware standards

Standard Organization	Networking Standards
ISO	• ISO 8877: Officially adopted the RJ connectors (e.g., RJ-11, RJ-45). • ISO 11801: Network cabling standard similar to EIA/TIA 568.
EIA/TIA	• TIA-568-C: Telecommunications cabling standards, used by nearly all voice, video, and data networks. • TIA-569-B: Commercial Building Standards for Telecommunications Pathways and Spaces. • TIA-598-C: Fiber optic color coding. • TIA-942: Telecommunications Infrastructure Standard for Data Centers.
ANSI	568-C: RJ-45 pinouts. Co-developed with EIA/TIA.
ITU-T	G.992: ADSL
IEEE	• 802.3: Ethernet • 802.11: Wireless LAN (WLAN) & Mesh (Wi-Fi certification) • 802.15: Bluetooth

Networking standards

Standard Organization	Networking Standards
IEEE	• 802.2: Logical Link Control (LLC) • 802.3: Ethernet • 802.4: Token bus • 802.5: Token ring • 802.11: Wireless LAN (WLAN) & Mesh (Wi-Fi certification) • 802.15: Bluetooth • 802.16: WiMax
ITU-T	• G.992: ADSL • G.8100 - G.8199: MPLS over Transport aspects • Q.921: ISDN • Q.922: Frame Relay
ISO	• HDLC (High Level Data Link Control) • ISO 9314: FDDI Media Access Control (MAC)
ANSI	• X3T9.5 and X3T12: Fiber Distributed Data Interface (FDDI)

IEEE 802 Standards

Standard	Name	Topic
802.1	Internetworking	Routing,Bridging,and network-to-network Communications
802.2	Logical Link Control	Error and flow control over data frames
802.3	Ethernet LAN	All forms of Ethernet media and interfaces
802.4	Token BUS LAN	All forms of Token Bus media and interfaces
802.5	Token Ring LAN	All forms of Token Ring media and interfaces
802.6	Metropolitan Area Network	MAN technologies,Addressing, and Services
802.7	Broadband technical Advisory Group	Broadband network media,interfaces, adn other Equipments
802.8	Fiber Optic Technical Advisory Group	Fiber Optic media used in token-passing Networks like FDDI
802.9	Integrated Voice/ Data Network	Integration of voice and data traffic Over a single network medium
802.10	Netwok Security	Network access controls,encryption,Certification, and other Security topics
802.11	Wireless Networks	Standards for wireless networking for many different broadcast frquencies and usage techniques
802.12	High-Speed Networking	A variety of 100 Mbps-plus technologies,including 100 BASE-VG
802.14	Cable Broadband LANs and MANs	Standards for designing network over coaxial cable-based broadband connections.
802.15	Wireless Personal Area Networks	The coexistence of wireless personal area networks with Others wireless devices in unlicensed frequency bands.
802.16	Broadband Wireless Access	The atmospheric interface and related functions associated with Wireless Local Loop(WLL)

OSI Model

Communication Architecture

- ❏ **Strategy for connecting host computers and other communicating equipment.**
- ❏ **Defines necessary elements for data communication between devices.**
- ❏ **A communication architecture, therefore, defines a standard for the communicating hosts.**
- ❏ **A programmer formats data in a manner defined by the communication architecture and passes it on to the communication software.**
- ❏ **Separating communication functions adds flexibility, for example, we do not need to modify the entire host software to include more communication devices.**

OSI Model

- OSI stands for Open System Interconnection is a reference model that describes how information from a software application in one computer moves through a physical medium to the software application in another computer.
- OSI consists of seven layers, and each layer performs a particular network function.
- OSI model was developed by the **International Organization for Standardization** (ISO) in 1984, and it is now considered as an architectural model for the inter-computer communications.
- OSI model divides the whole task into seven smaller and manageable tasks. Each layer is assigned a particular task.
- Each layer is self-contained, so that task assigned to each layer can be performed independently.

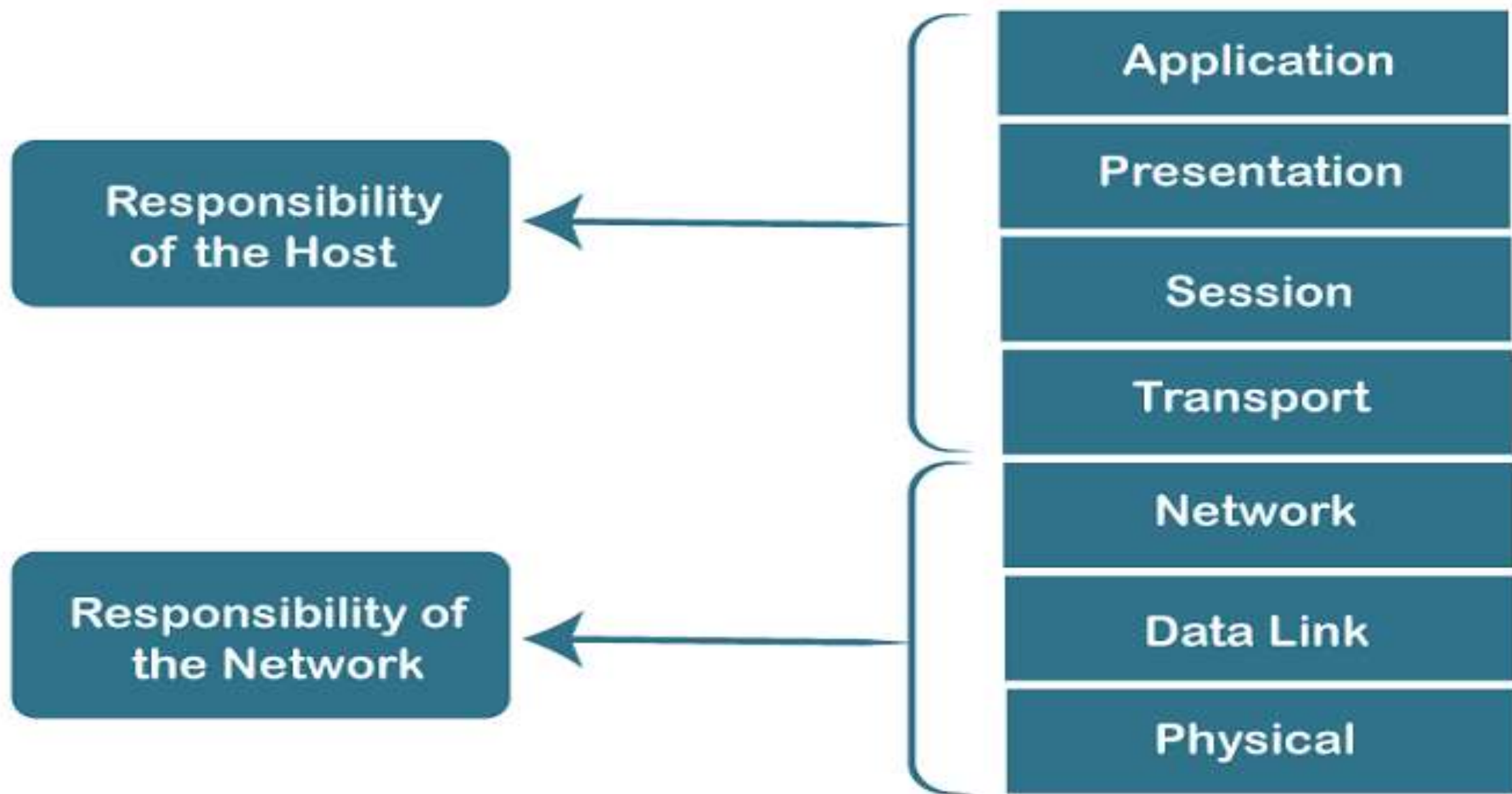
Layered architecture

- **Layer architecture simplifies the network design.**
 - **It is easy to debug network applications in a layered architecture network.**
 - **The network management is easier due to the layered architecture.**
 - **Network layers follow a set of rules, called protocol.**
 - **The protocol defines the format of the data being exchanged, and the control and timing for the handshake between layers.**
- 

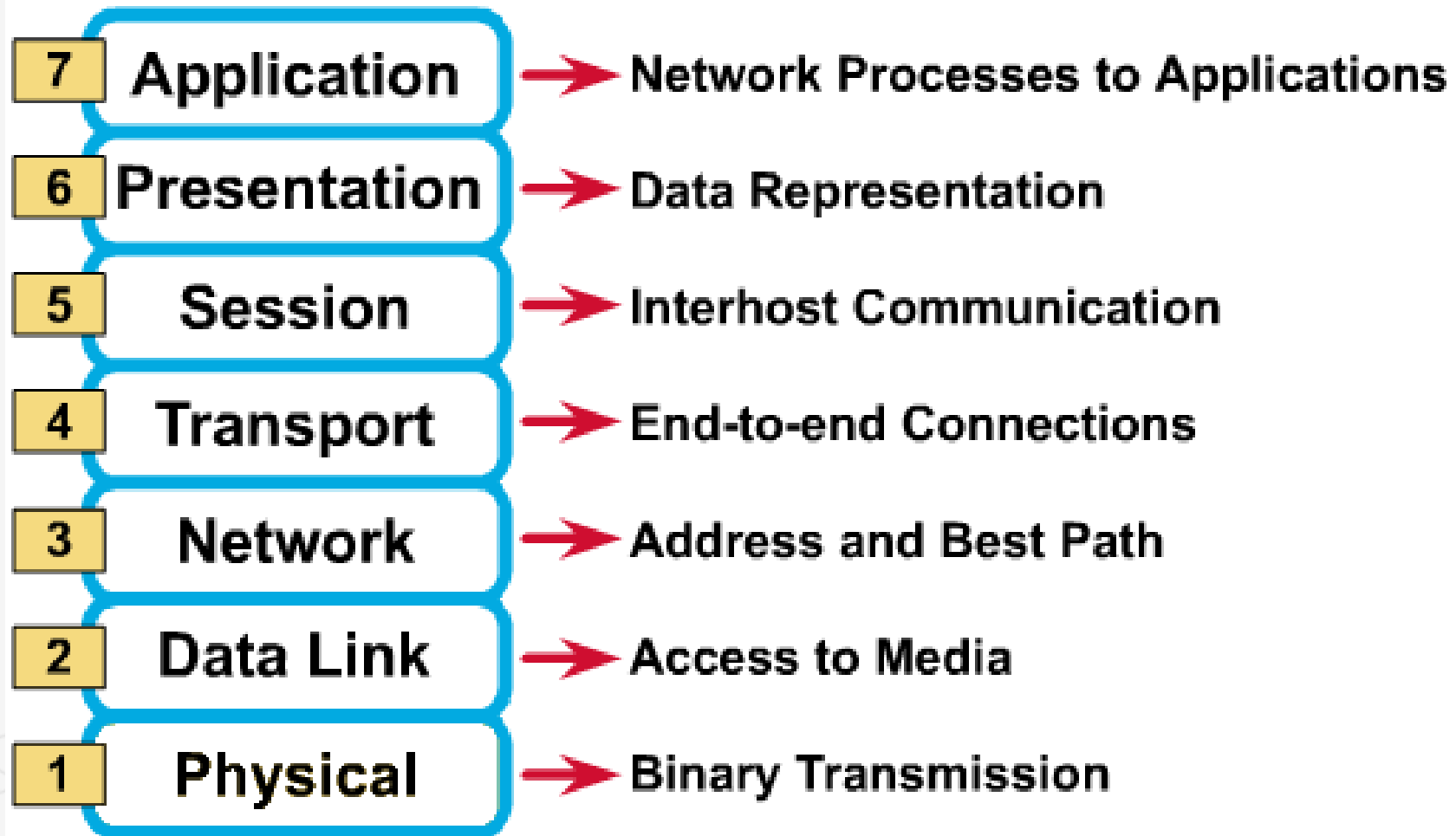
OSI Reference Model

- ❏ **The OSI model is now considered the primary Architectural model for inter-computer communications.**
- ❏ **The OSI model describes how information or data makes its way from application programmes (such as spreadsheets) through a network medium (such as wire) to another application programme located on another network.**
- ❏ **The OSI reference model divides the problem of moving information between computers over a network medium into SEVEN smaller and more manageable problems .**
- ❏ **This separation into smaller more manageable functions is known as layering.**

Characteristics of OSI Model



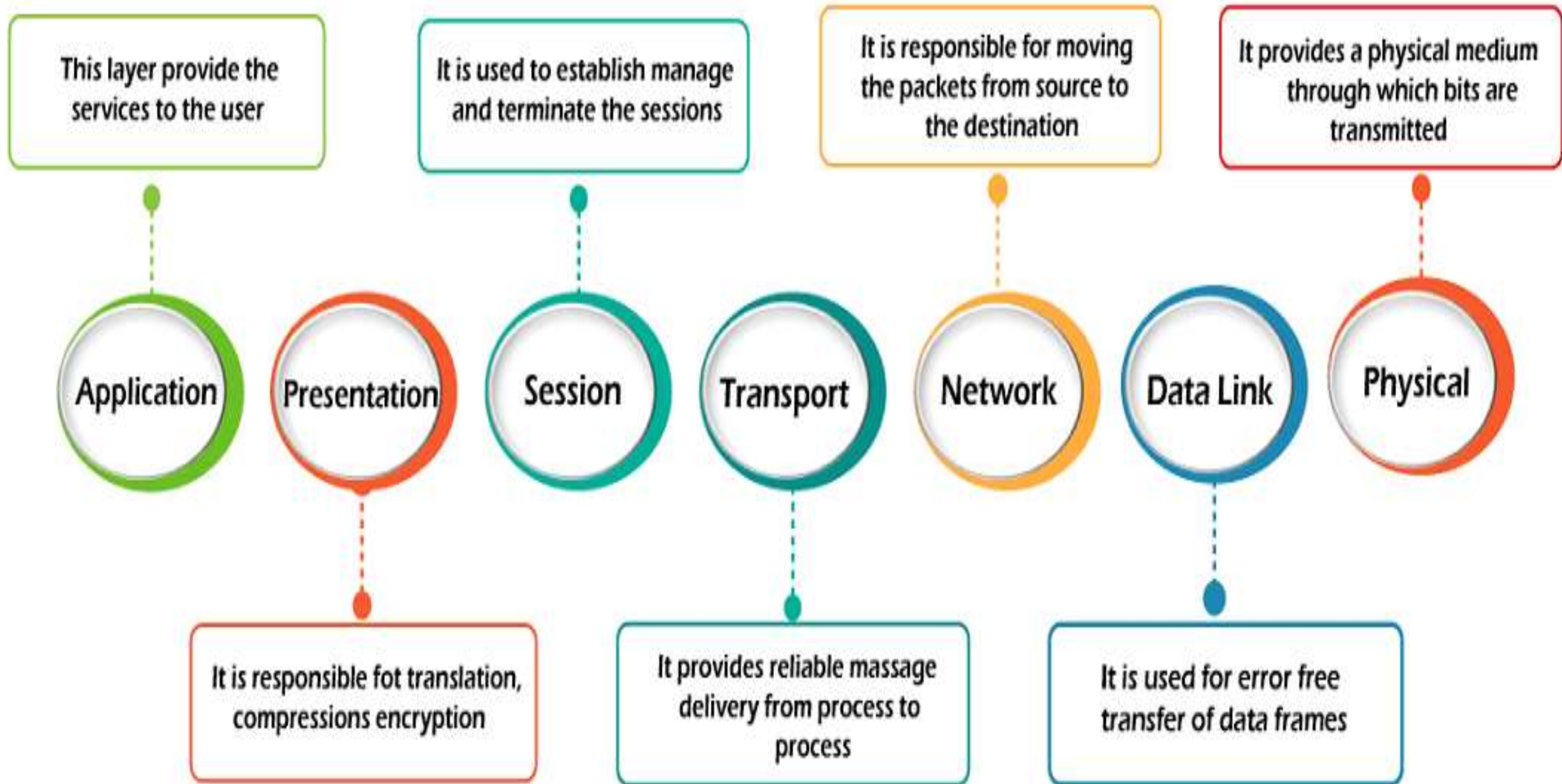
OSI Reference Model: 7 Layers

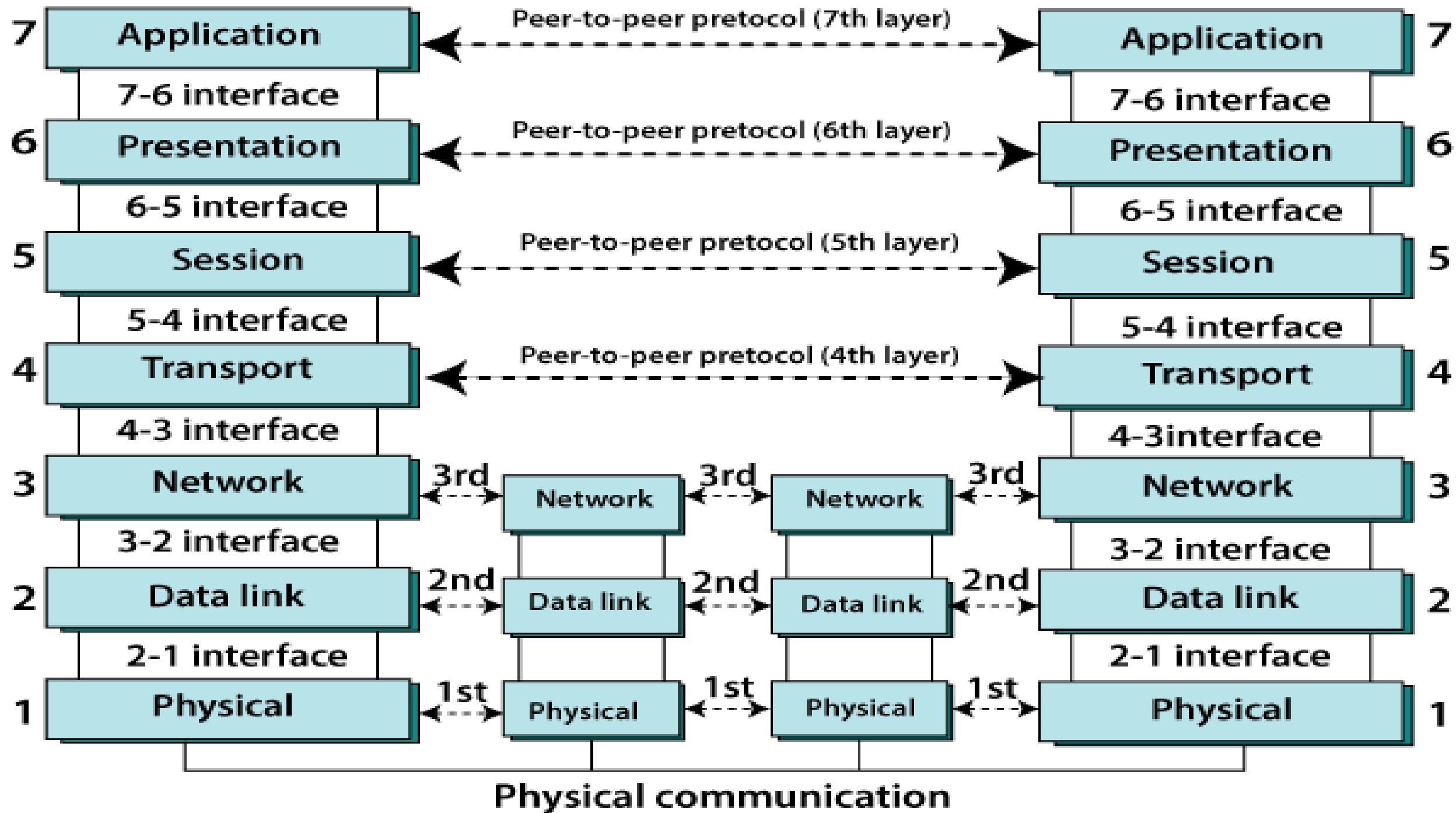


The OSI model is divided into two layers: upper layers and lower layers.

- The upper layer of the OSI model mainly deals with the application related issues, and they are implemented only in the software.
- The application layer is closest to the end user.
- Both the end user and the application layer interact with the software applications.
- An upper layer refers to the layer just above another layer.
- The lower layer of the OSI model deals with the data transport issues.
- The data link layer and the physical layer are implemented in hardware and software.
- The physical layer is the lowest layer of the OSI model and is closest to the physical medium.
- The physical layer is mainly responsible for placing the information on the physical medium.

7 Layers of OSI Model

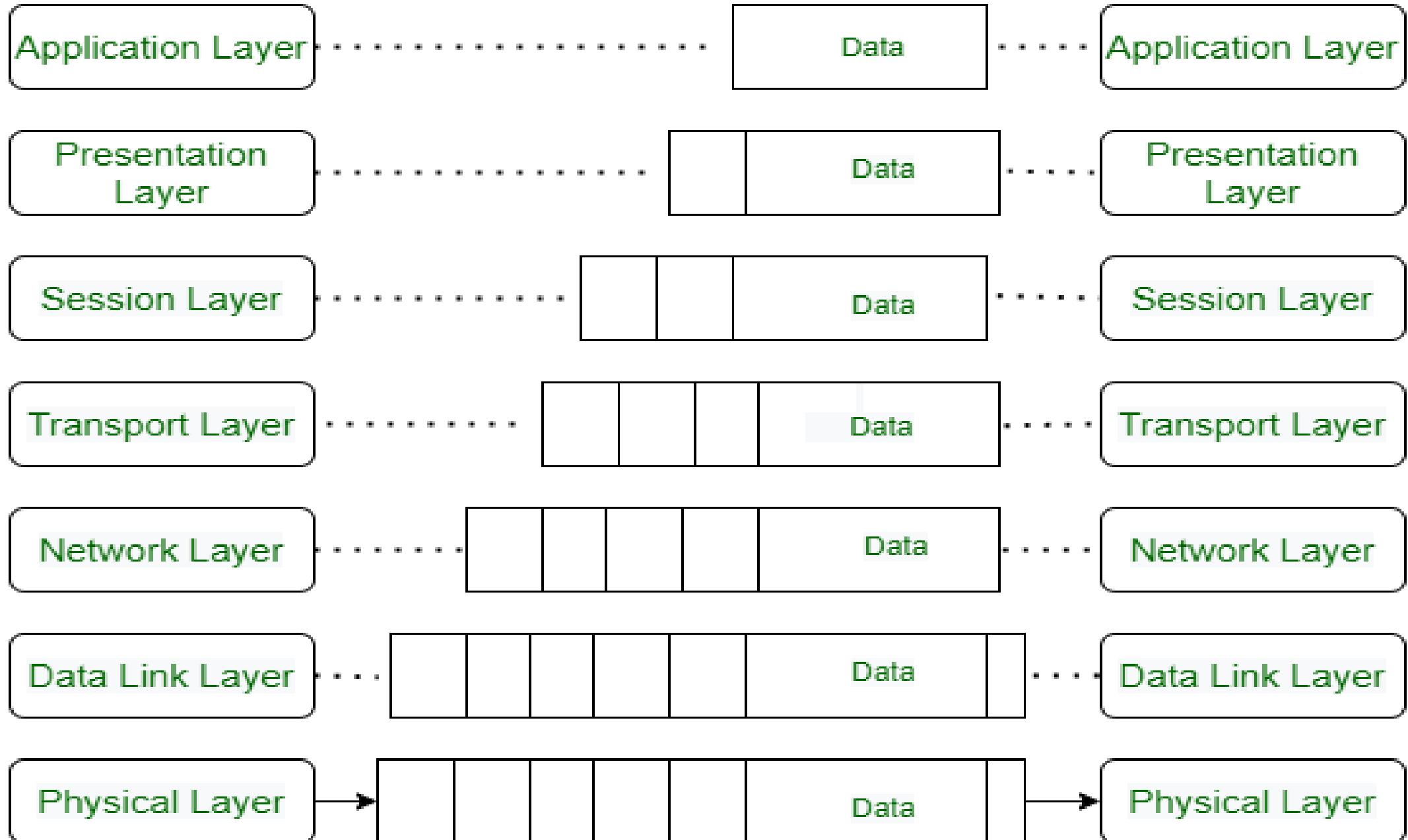




Sender

Build-up packets

Reciever



Physical layer



Note:

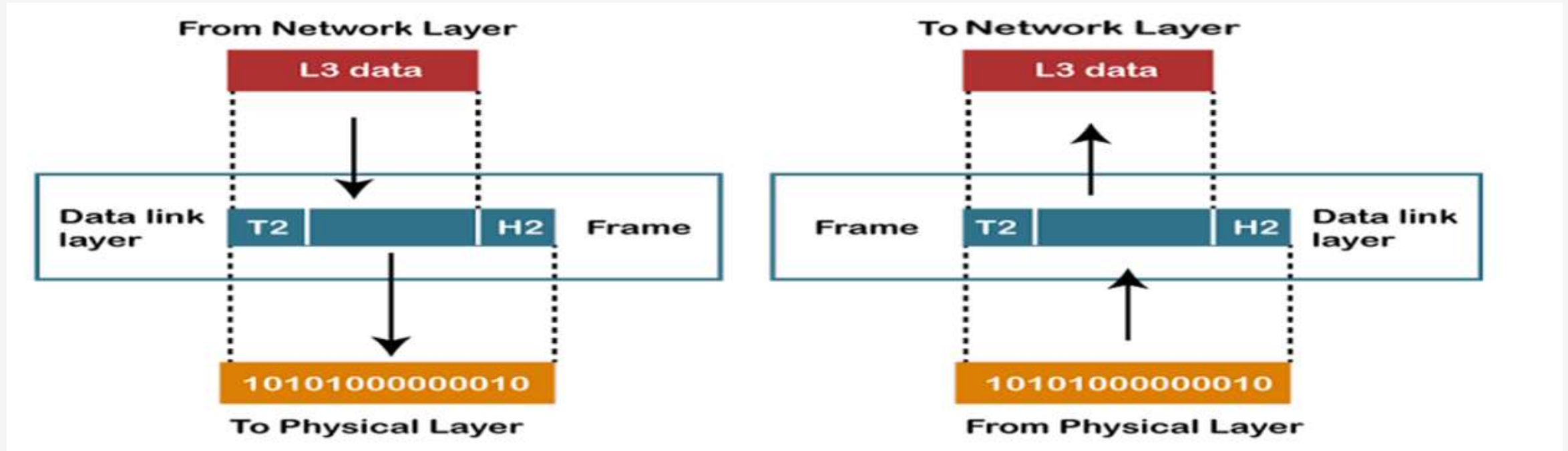
- ✓ Hub, Repeater, Modem, and Cables are Physical Layer devices.
- ✓ Network Layer, Data Link Layer, and Physical Layer are also known as Lower Layers or Hardware Layers.

- The lowest layer of the OSI reference model is the physical layer.
- It is responsible for the actual physical connection between the devices.
- The physical layer contains information in the form of bits.
- It is responsible for transmitting individual bits from one node to the next.
- When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.

Functions of the Physical Layer

- **Bit synchronization:** The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- **Bit rate control:** The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- **Physical topologies:** Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus, star, or mesh topology.
- **Transmission mode:** Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are Simplex, half-duplex and full-duplex.

Data-Link Layer



- The data link layer is responsible for the node-to-node delivery of the message.
- The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer.
- When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address.

The Data Link Layer is divided into two sub-layers:

Logical Link Control Layer

- ✓ It is responsible for transferring the packets to the Network layer of the receiver that is receiving.
- ✓ It identifies the address of the network layer protocol from the header.
- ✓ It also provides flow control.

Media Access Control Layer

- ✓ A Media access control layer is a link between the Logical Link Control layer and the network's physical layer.
- ✓ It is used for transferring the packets over the network.

- The packet received from the Network layer is further divided into frames depending on the frame size of the NIC(Network Interface Card).
-
- DLL also encapsulates Sender and Receiver's MAC address in the header.
 - The Receiver's MAC address is obtained by placing an ARP(Address Resolution Protocol) request onto the wire asking "Who has that IP address?" and the destination host will reply with its MAC address.



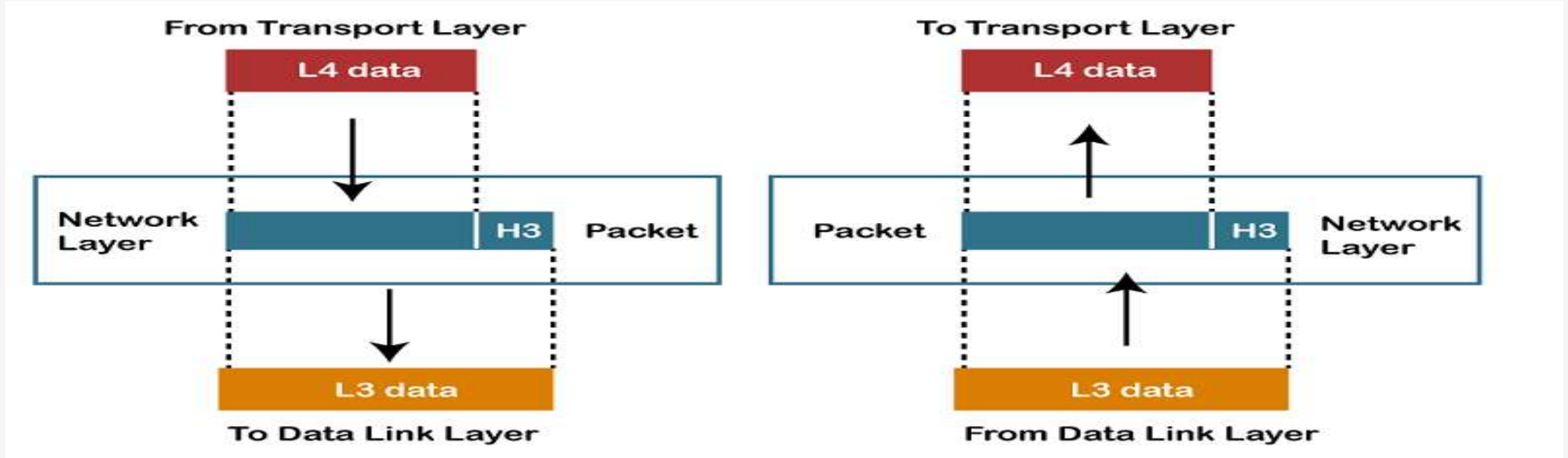
Note:

- Packet in the Data Link layer is referred to as Frame.
- Data Link layer is handled by the NIC (Network Interface Card) and device drivers of host machines.
- Switch & Bridge are Data Link Layer devices.

Functions of the Data Link Layer

- **Framing:** Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.
- **Physical addressing:** After creating frames, the Data link layer adds physical addresses (MAC addresses) of the sender and/or receiver in the header of each frame.
- **Error control:** The data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.
- **Flow Control:** The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- **Access control:** When a single communication channel is shared by multiple devices, the MAC sub-layer of the data link layer helps to determine which device has control over the channel at a given time.

Network Layer



- The network layer works for the transmission of data from one host to the other located in different networks.
- It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available.
- The sender & receiver's IP addresses are placed in the header by the network layer.

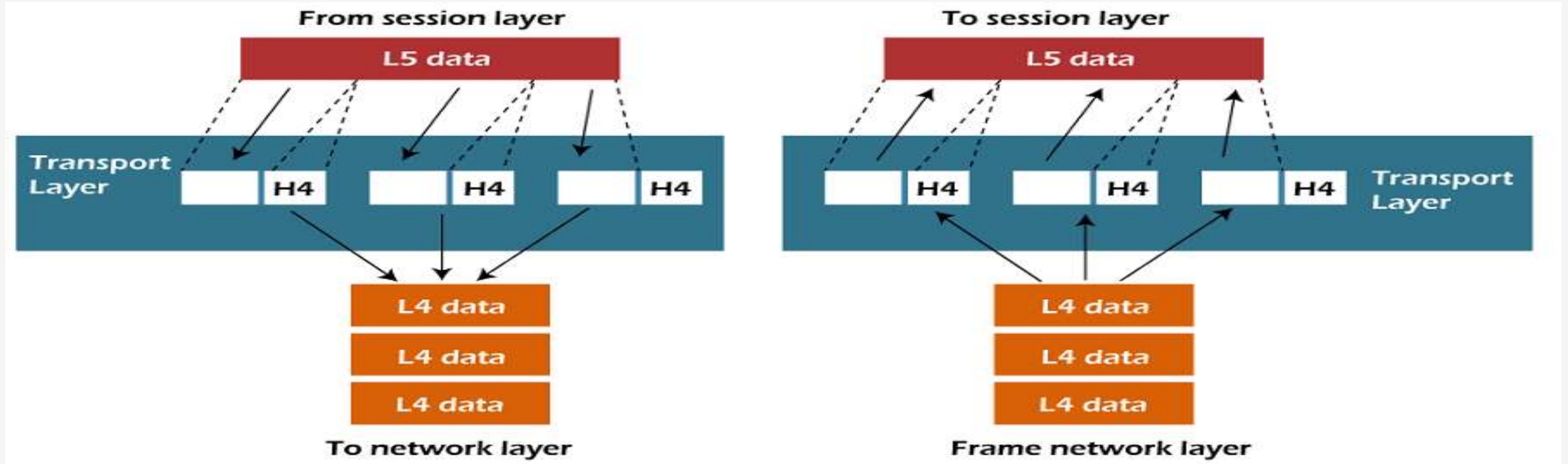
Functions of the Network Layer

- **Internetworking:** An internetworking is the main responsibility of the network layer. It provides a logical connection between different devices.
- **Addressing:** A Network layer adds the source and destination address to the header of the frame. Addressing is used to identify the device on the internet.
- **Routing:** Routing is the major component of the network layer, and it determines the best optimal path out of the multiple paths from source to the destination.
- **Packetizing:** A Network Layer receives the packets from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).

Note:

- ✓ Segment in the Network layer is referred to as Packet.
- ✓ Network layer is implemented by networking devices such as routers and switches.

Transport Layer



- The transport layer provides services to the application layer and takes services from the network layer.
- The data in the transport layer is referred to as Segments.
- It is responsible for the End to End Delivery of the complete message.
- The transport layer also provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found.
- At the sender's side: The transport layer receives the formatted data from the upper layers, performs Segmentation, and also implements Flow & Error control to ensure proper data transmission.
- It also adds Source and Destination port numbers in its header and forwards the segmented data to the Network Layer.

The two protocols used in this layer are:

Transmission Control Protocol

- It is a standard protocol that allows the systems to communicate over the internet.
- It establishes and maintains a connection between hosts.
- When data is sent over the TCP connection, then the TCP protocol divides the data into smaller units known as segments.
- Each segment travels over the internet using multiple routes, and they arrive in different orders at the destination.
- The transmission control protocol reorders the packets in the correct order at the receiving end.

User Datagram Protocol

- User Datagram Protocol is a transport layer protocol.
- It is an unreliable transport protocol as in this case receiver does not send any acknowledgment when the packet is received, the sender does not wait for any acknowledgment. Therefore, this makes a protocol unreliable.

Functions of the Transport Layer

- **Segmentation and Reassembly:** This layer accepts the message from the (session) layer, and breaks the message into smaller units. Each of the segments produced has a header associated with it. The transport layer at the destination station reassembles the message.
- **Service Point Addressing:** To deliver the message to the correct process, the transport layer header includes a type of address called service point address or port address. Thus by specifying this address, the transport layer makes sure that the message is delivered to the correct process.

Services Provided by Transport Layer

Connection-Oriented Service

Connectionless Service

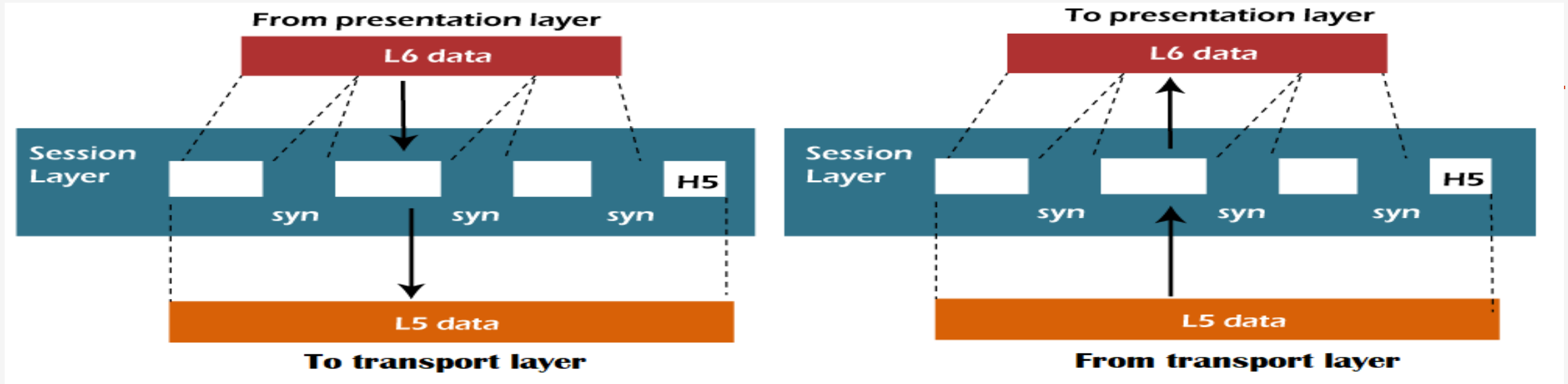
1. **Connection-Oriented Service:** It is a three-phase process that includes
 - Connection Establishment
 - Data Transfer
 - Termination/disconnection

In this type of transmission, the receiving device sends an acknowledgment, back to the source after a packet or group of packets is received. This type of transmission is reliable and secure.

2. **Connectionless service:** It is a one-phase process and includes Data Transfer. In this type of transmission, the receiver does not acknowledge receipt of a packet. This approach allows for much faster communication between devices. Connection-oriented service is more reliable than connectionless Service.

-
- Data in the Transport Layer is called Segments.
 - Transport layer is operated by the Operating System.
 - It is a part of the OS and communicates with the Application Layer by making system calls.
 - The transport layer is called as Heart of the OSI model.
 - Device or Protocol Use : TCP, UDP NetBIOS, PPTP

Session Layer

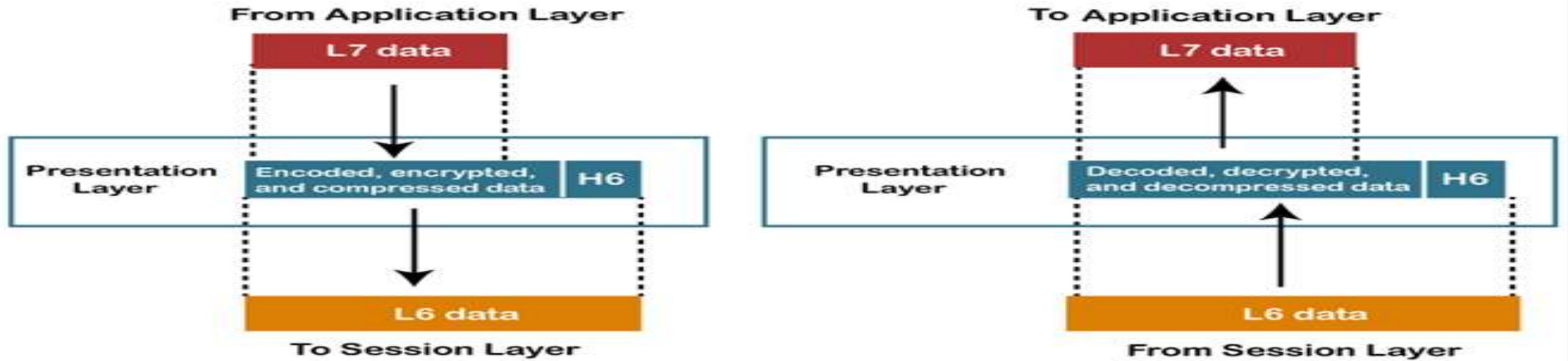


This layer is responsible for the establishment of connection, maintenance of sessions, and authentication, and also ensures security.

Functions of the Session Layer

- **Session establishment, maintenance, and termination:** The layer allows the two processes to establish, use and terminate a connection.
- **Synchronization:** This layer allows a process to add checkpoints that are considered synchronization points in the data. These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely and data loss is avoided.
- **Dialog Controller:** The session layer allows two systems to start communication with each other in half-duplex or full-duplex.

Presentation Layer



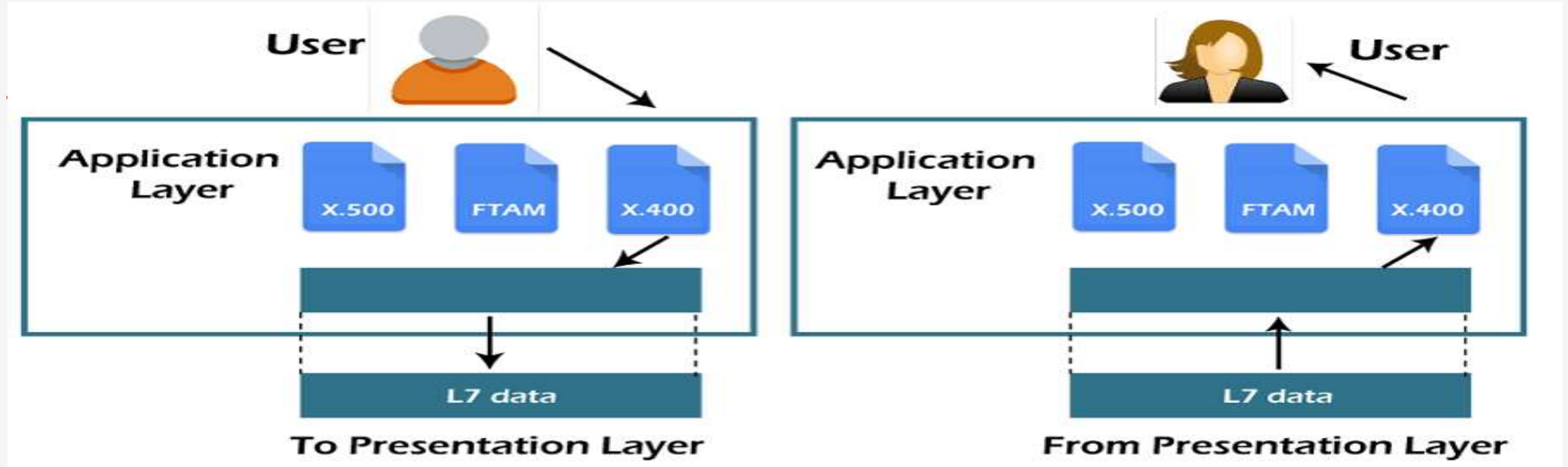
- The presentation layer is also called the Translation layer.
- The data from the application layer is extracted here and manipulated as per the required format to transmit over the network.

Functions of the Presentation Layer

- **Translation:** For example, ASCII to EBCDIC.
- **Encryption/ Decryption:** Data encryption translates the data into another form or code. The encrypted data is known as the cipher text and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- **Compression:** Reduces the number of bits that need to be transmitted on the network.

Note: Device or Protocol Use : JPEG, MPEG, GIF

Application Layer



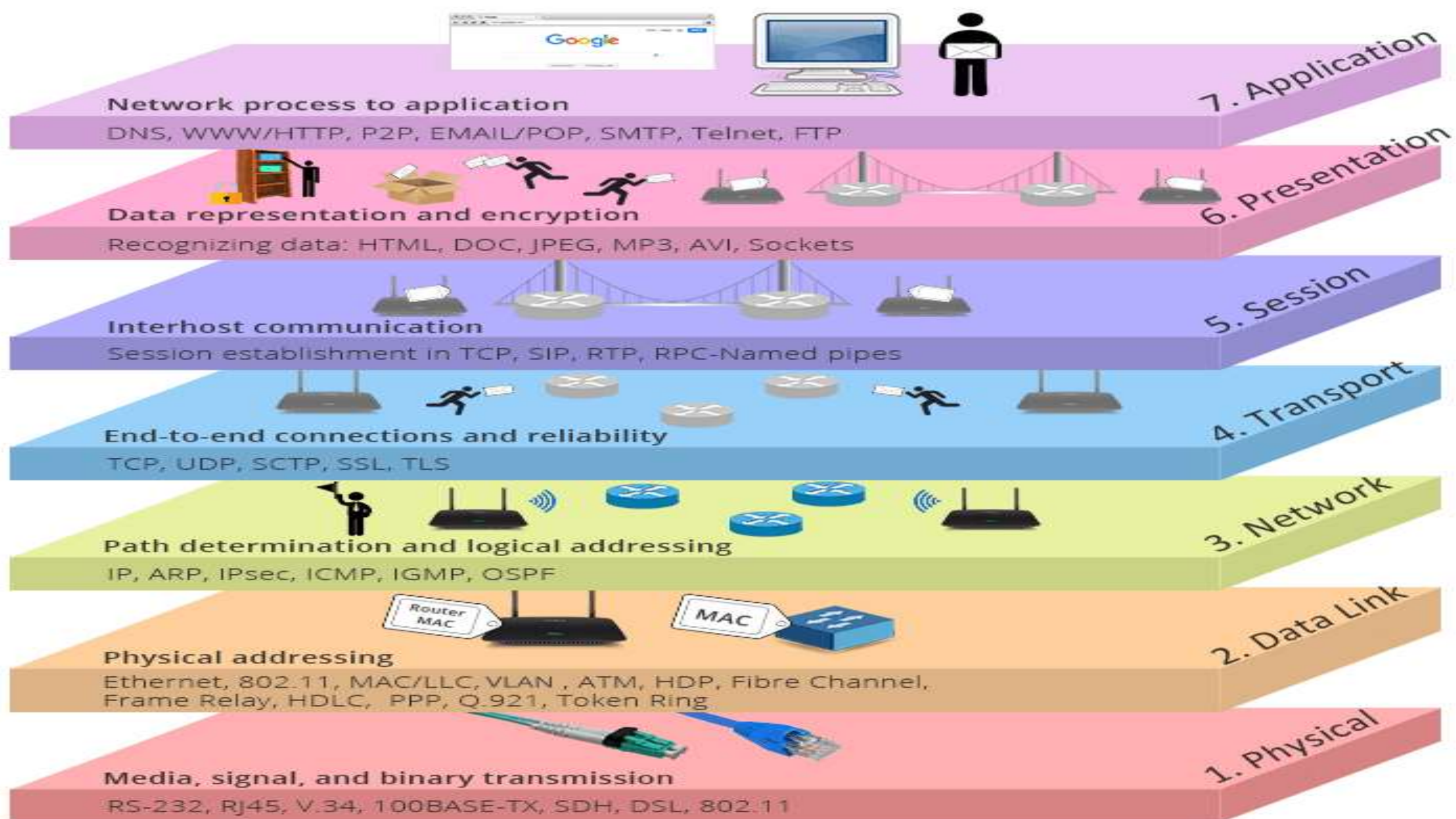
- Top Layer of the OSI Reference Model, we find the Application layer which is implemented by the network applications.
- These applications produce the data, which has to be transferred over the network.
- This layer also serves as a window for the application services to access the network and for displaying the received information to the user.

Example: Application – Browsers, Skype Messenger, etc.

Functions of Application layer:

- **File transfer, access, and management (FTAM):** An application layer allows a user to access the files in a remote computer, to retrieve the files from a computer and to manage the files in a remote computer.
- **Mail services:** An application layer provides the facility for email forwarding and storage.
- **Directory services:** An application provides the distributed database sources and is used to provide that global information about various objects.

Layer No	Layer Name	Responsibility	Information Form(Data Unit)	Device or Protocol
7	Application Layer	Helps in identifying the client and synchronizing communication.	Message	SMTP
6	Presentation Layer	Data from the application layer is extracted and manipulated in the required format for transmission.	Message	JPEG, MPEG, GIF
5	Session Layer	Establishes Connection, Maintenance, Ensures Authentication, and Ensures security.	Message	Gateway
4	Transport Layer	Take Service from Network Layer and provide it to the Application Layer.	Segment	Firewall
3	Network Layer	Transmission of data from one host to another, located in different networks.	Packet	Router
2	Data Link Layer	Node to Node Delivery of Message.	Frame	Switch, Bridge
1	Physical Layer	Establishing Physical Connections between Devices.	Bits	Hub, Repeater, Modem, Cables



TCP/IP model

- TCP/IP was designed and developed by the Department of Defense (DoD) in the 1960s and is based on standard protocols.
- It stands for Transmission Control Protocol/Internet Protocol.
- The TCP/IP model is a concise version of the OSI model.
- It contains four layers, unlike the seven layers in the OSI model.
- The number of layers is sometimes referred to as five or four.
- The Physical Layer and Data Link Layer are referred to as one single layer as the 'Physical Layer' or 'Network Interface / Access Layer' in the 4-layer reference.

What Is the TCP/IP Protocol Suite?

The TCP/IP protocol suite (Transmission Control Protocol/Internet Protocol) is a comprehensive set of communication protocols that governs how data is packetized, addressed, transmitted, routed, and reassembled across networks.

It provides the foundational rules that allow billions of devices — smartphones, laptops, servers, IoT hardware — to communicate over both the public internet and private networks.

Without TCP/IP, there is no internet. Every email you send, every website you visit, every video call you join depends on this protocol stack working reliably in the background.



THE SUITE DOES FOUR THINGS

Breaks data into packets — Large chunks of data are split into manageable units called data packets, each tagged with source and destination information.

Addresses each packet — Using **IP addresses**, every packet is labeled so networks know where it came from and where it's going.

Routes packets across networks — Routers and gateways forward packets across multiple network boundaries until they reach the destination.

Reassembles data at the destination — The receiving device collects all packets and reconstructs the original data in the correct order.

Key Highlights

TCP/IP is a suite of communication protocols — not a single protocol.

It defines how data is packetized, addressed, transmitted, routed, and reassembled.

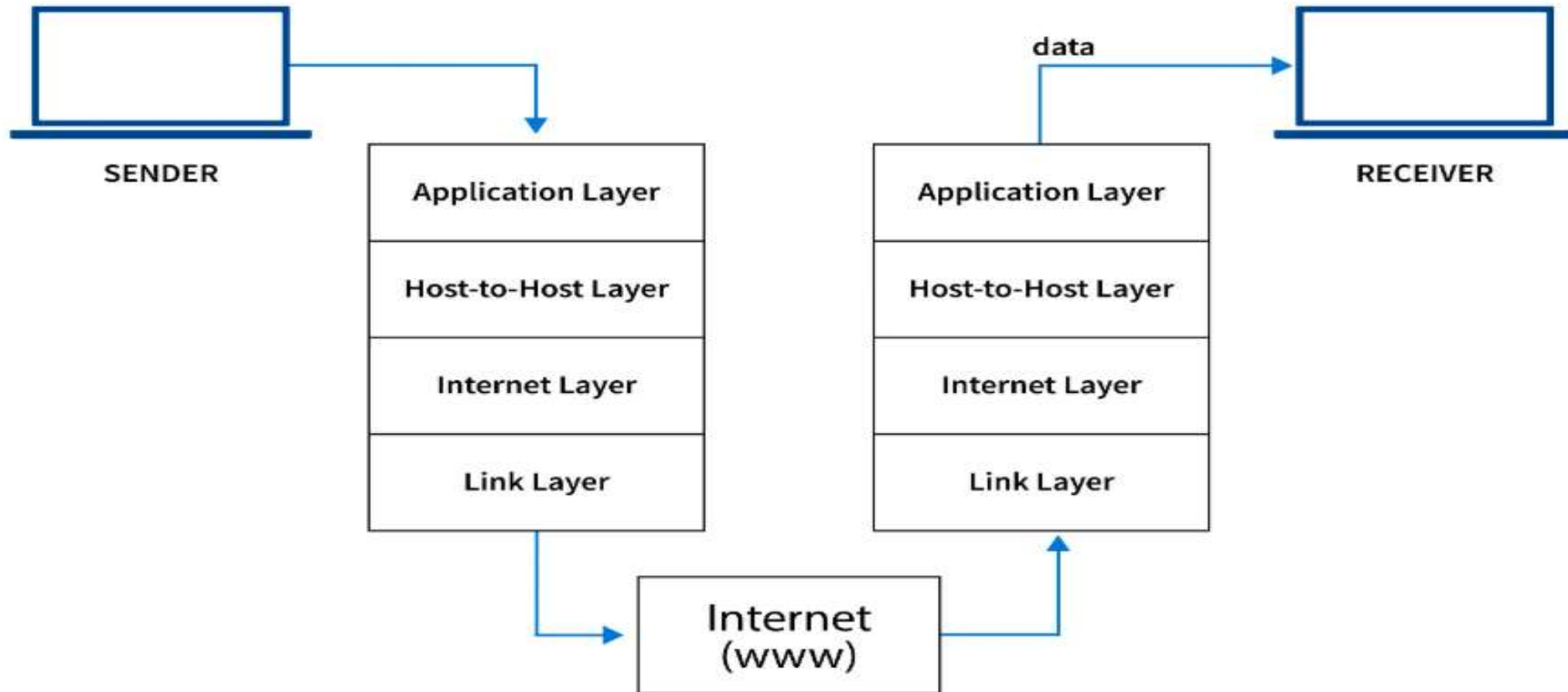
It is the practical, implementation-based model used in real-world networks, unlike the OSI model, which is primarily theoretical.

It was developed by DARPA in the 1970s, designed by Vinton Cerf and Robert Kahn, and adopted by ARPANET on January 1, 1983.

It supports communication across heterogeneous networks — different hardware, different operating systems, different vendors — all speaking the same protocol language.

Layer	Primary Responsibility
Application Layer	Interfaces with user applications; handles data formatting and communication methods
Transport Layer (Host-to-Host)	Ensures end-to-end communication, reliability, and flow control
Internet Layer (Network Layer)	Handles logical addressing, routing, and packet forwarding across networks
Network Access Layer (Link Layer)	Manages physical transmission over the network medium

How Does the TCP/IP Protocol Work?



Step 1 — Data Creation at the Application Layer

A user creates data — typing a message, requesting a webpage, uploading a file. The application (a browser, email client, or file transfer tool) prepares this data using an **application layer protocol** such as:

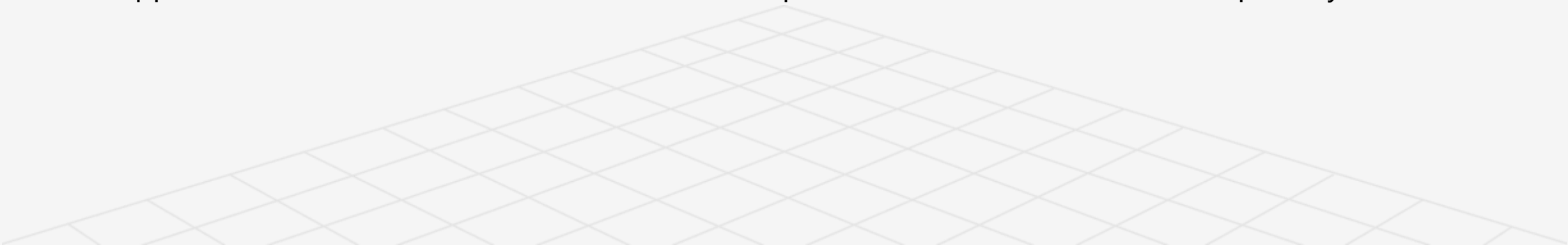
HTTP/HTTPS (Hypertext Transfer Protocol) — for web pages

SMTP (Simple Mail Transfer Protocol) — for sending email

FTP (File Transfer Protocol) — for transferring files

DNS (Domain Name System) — for resolving domain names to IP addresses

The application adds its own header information and passes the data down to the transport layer.



Step 2 — Segmentation at the Transport Layer

The **transport layer protocols** — TCP or UDP — take the application data and segment it into smaller units:

TCP adds sequence numbers, port numbers, and checksums. It establishes a connection with the receiver using a three-way handshake (SYN → SYN-ACK → ACK). This ensures both sides are ready before any data flows.

UDP skips the handshake entirely. It adds port numbers and a minimal checksum, then sends the data immediately. No guarantees of delivery, order, or error recovery.

The choice between TCP and UDP depends on the application. A file download needs TCP's reliability. A live video call needs UDP's speed.



Step 3 — Addressing and Routing at the Internet Layer

The **internet layer** takes each transport segment and encapsulates it into an **IP datagram**. The IP header contains:

Source IP address — Where the data is coming from.

Destination IP address — Where the data is going.

TTL (Time to Live) — A counter that decreases at each router hop, preventing packets from circulating forever.

Protocol field — Identifying whether the payload is TCP or UDP.

An IP router examines the destination network address in each datagram and forwards it toward the destination network. In large networks, routing protocols like OSPF, BGP, and EIGRP help routers exchange path information and determine optimal routes.



Step 4 — Physical Transmission at the Network Access Layer

The network access layer (also called the **network interface layer** or **link layer**) converts the IP datagram into a frame suitable for the **physical network**. This layer:

Maps IP addresses to hardware (MAC) addresses using **ARP (Address Resolution Protocol)**.

Encapsulates data into frames with source and destination MAC addresses.

Transmits the frame over the **network medium** — Ethernet cables, fiber optics, Wi-Fi radio signals, or serial links using the **Point-to-Point Protocol (PPP)**.

Handles **error detection** at the frame level.



Step 5 — Reception and Reassembly

At the receiving device, the entire process runs in reverse:

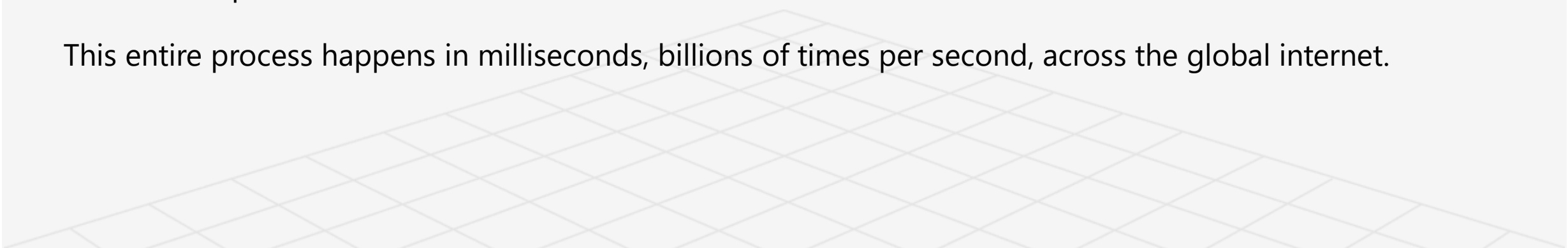
The **network access layer** receives the frame, checks for errors, and extracts the IP datagram.

The **internet layer** reads the destination IP address, confirms it matches the device, and removes the IP header.

The **transport layer** reassembles the segments in order (TCP) or delivers them as they arrive (UDP), checking for errors and requesting retransmission if needed.

The **application layer** presents the reconstructed data to the user — a rendered webpage, a delivered email, a completed file.

This entire process happens in milliseconds, billions of times per second, across the global internet.



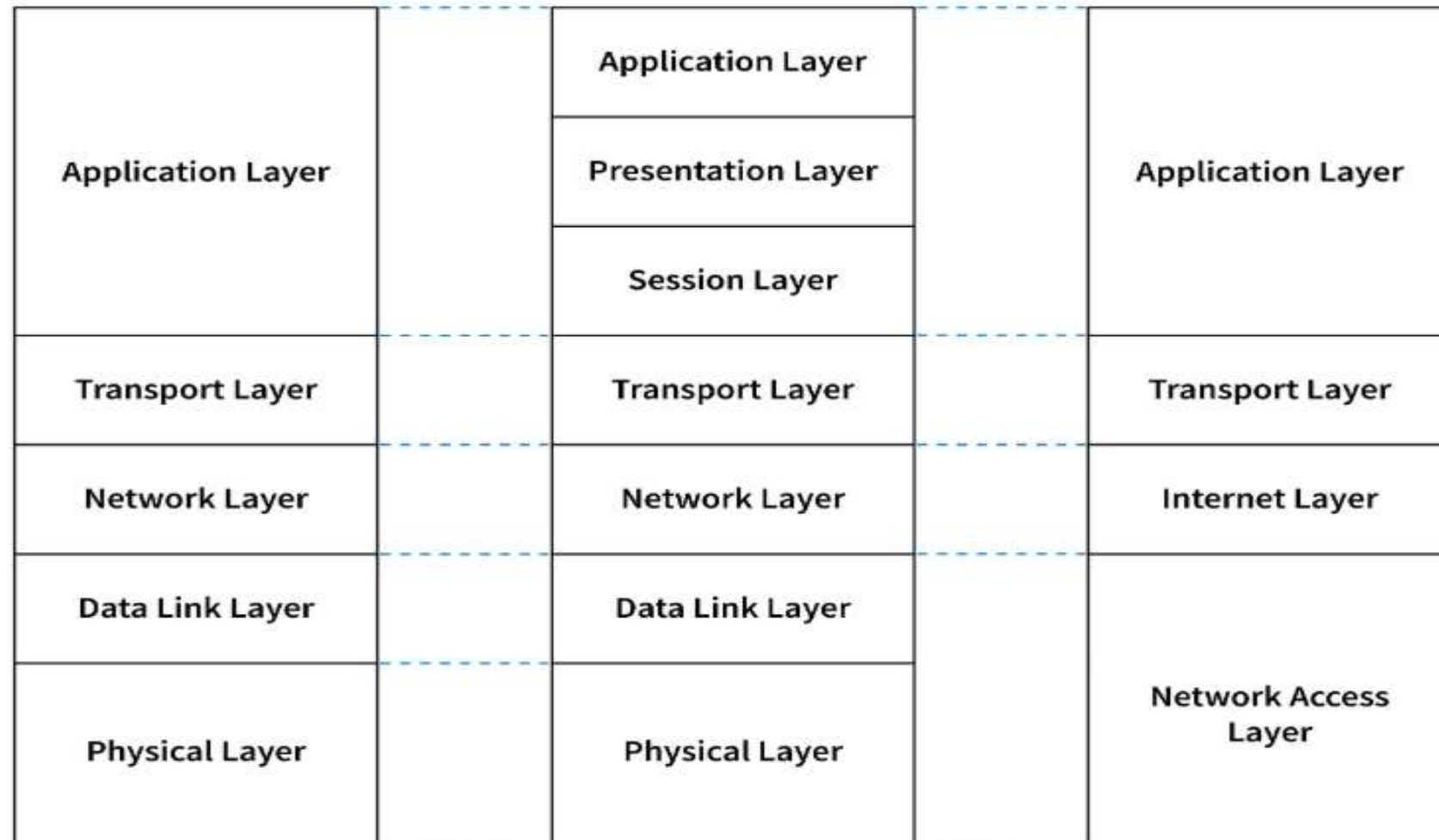
TCP/IP's success comes down to three principles

Standardization — Every device follows the same protocol rules, regardless of vendor or platform.

Layered abstraction — Each layer handles its own responsibility without needing to understand the others. The application layer doesn't need to know whether the physical connection is Ethernet or Wi-Fi.

Scalability — The same protocol suite handles communication on a home network with three devices and the global internet with billions.





5 - Layer TCP/IP

OSI

4 - Layer